

# Duality, Wasserstein-1, and Semi-discrete OT

Dual potentials, Lipschitz witnesses, Laguerre cells, and quantization algorithms.

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Optimal Transport for Machine Learners

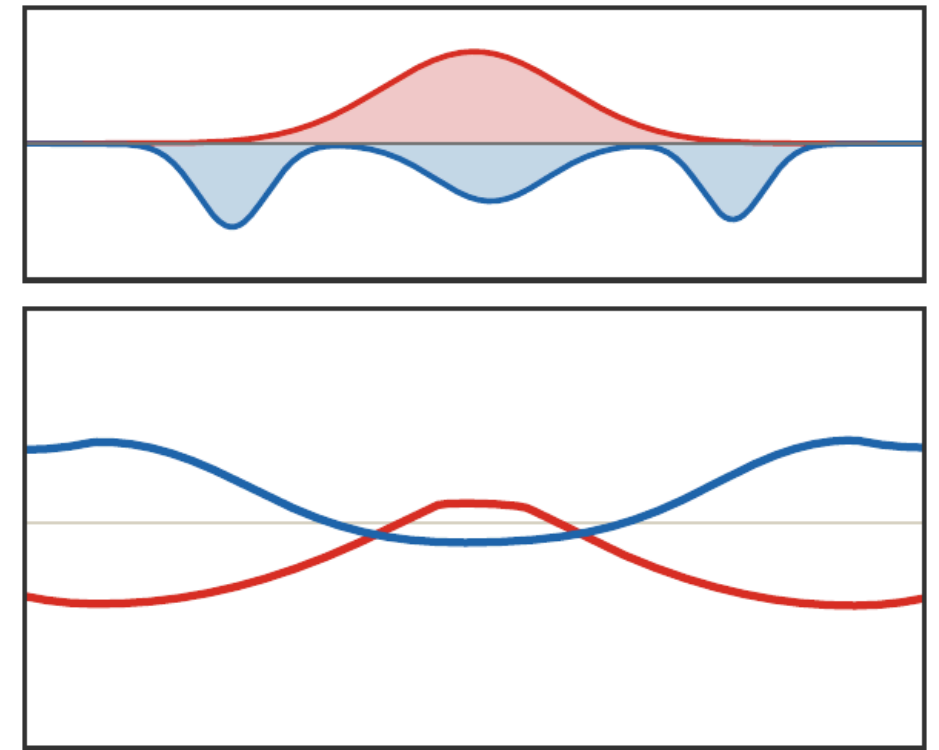
# Why duality matters

Kantorovich relaxation turns OT into a linear program. Its dual exposes functions rather than couplings.

This is useful because dual potentials:

- certify optimality,
- define gradients of OT losses,
- produce  $W_1$  witnesses and GAN critics,
- drive semi-discrete algorithms.

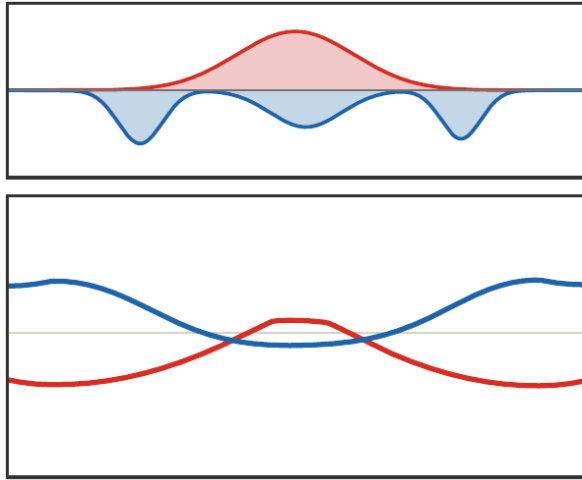
**source:**  $\alpha, x, a$  **target:**  $\beta, y, b$  **plan/map:**  $\pi, P, T$



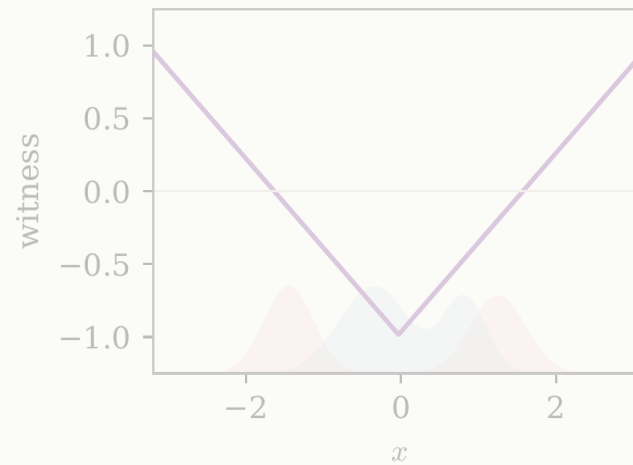
Dual potentials touch the cost on transported pairs.

# Roadmap: duality

## 1. Duality



## 2. Wasserstein-1 and IPMs



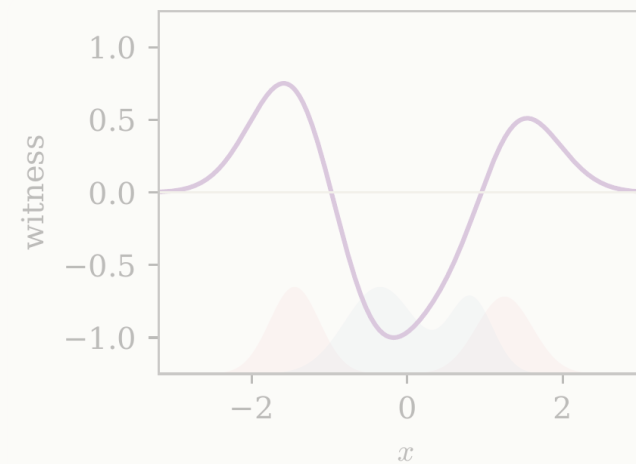
## 3. Semi-discrete OT



## 4. Quantization



## 5. Learning links



# Discrete primal and dual

## Discrete Kantorovich primal.

For  $a \in \Sigma_n$ ,  $b \in \Sigma_m$  and  $C \in \mathbb{R}^{n \times m}$ ,

$$\mathcal{L}_C(a, b) = \min_{P \geq 0} \langle C, P \rangle \quad \text{s.t.} \quad P 1_m = a, \quad P^\top 1_n = b.$$

## Discrete dual.

$$\mathcal{L}_C(a, b) = \max_{f \in \mathbb{R}^n, g \in \mathbb{R}^m} \langle f, a \rangle + \langle g, b \rangle \quad \text{s.t.} \quad f_i + g_j \leq C_{i,j}.$$

# Lagrangian derivation

The Lagrangian of the primal is

$$\langle C, P \rangle + \langle f, a - P \mathbf{1}_m \rangle + \langle g, b - P^\top \mathbf{1}_n \rangle.$$

Minimizing over  $P \geq 0$  gives a finite value if and only if

$$C_{i,j} - f_i - g_j \geq 0 \quad \forall i, j.$$

This yields the dual constraints  $f_i + g_j \leq C_{i,j}$ .

# Complementary slackness

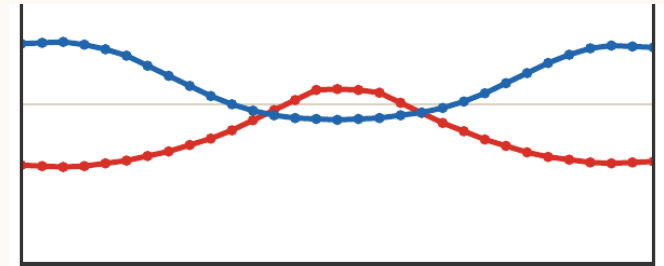
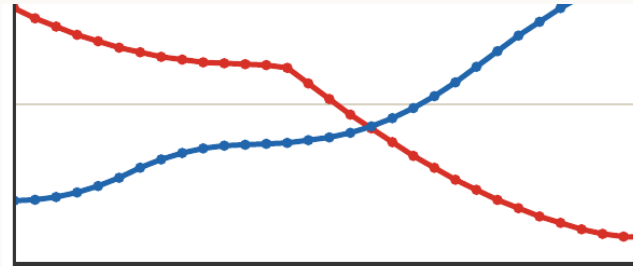
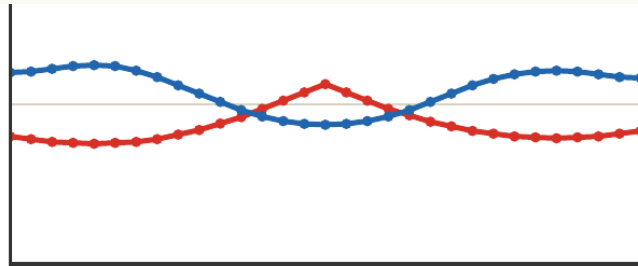
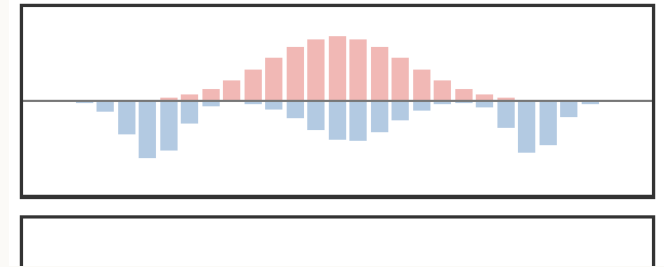
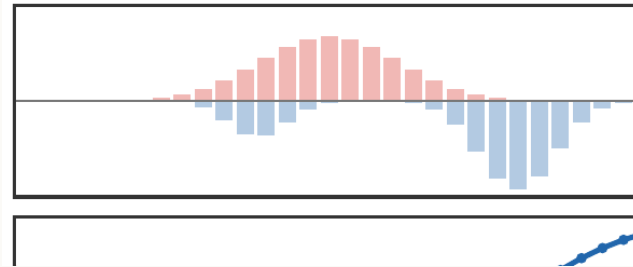
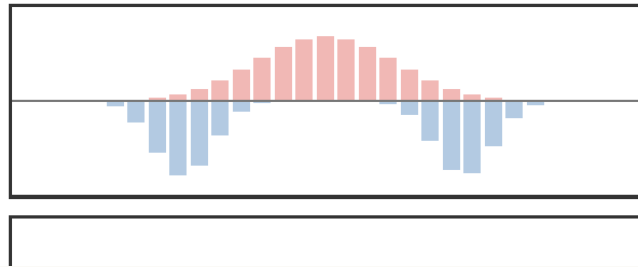
## Primal-dual contact.

If  $P$  and  $(f, g)$  are optimal, then

$$P_{i,j} > 0 \implies f_i + g_j = C_{i,j}.$$

# Discrete dual witnesses

Dual variables are not just algebraic multipliers: they are certificates explaining which transport edges can be active.



# Live panel: discrete dual potentials



Dual certificates identify the active transport pairs.



# Continuous duality

## Kantorovich duality.

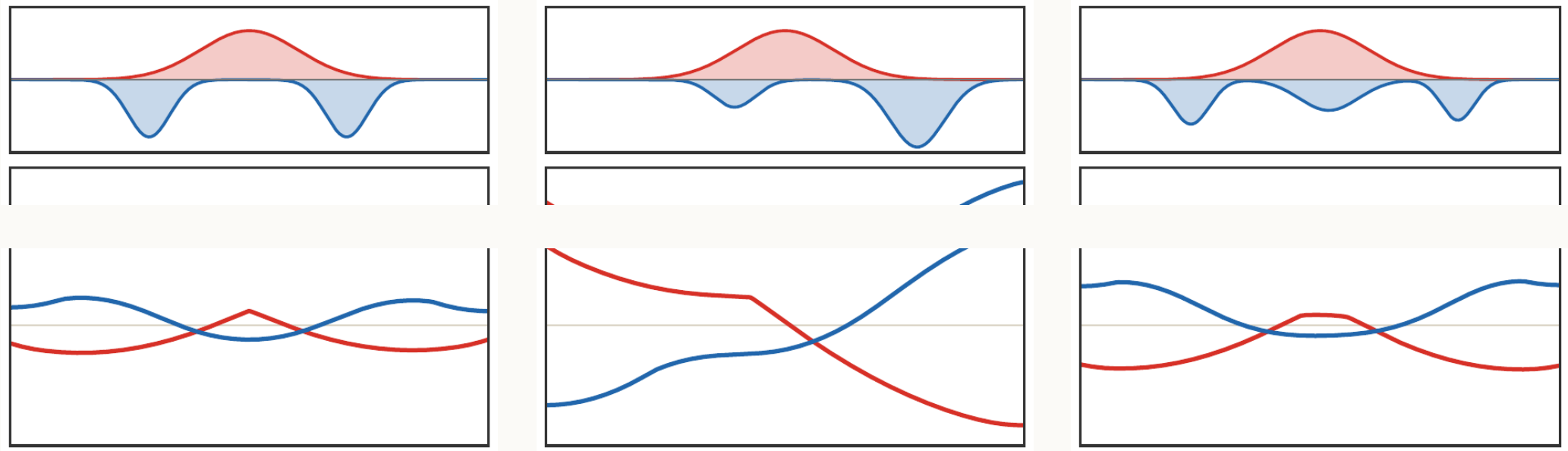
On Polish spaces, for a lower-semicontinuous cost  $c$  bounded from below,

$$\mathcal{L}_c(\alpha, \beta) = \sup_{f, g} \int f d\alpha + \int g d\beta \quad \text{s.t.} \quad f(x) + g(y) \leq c(x, y).$$

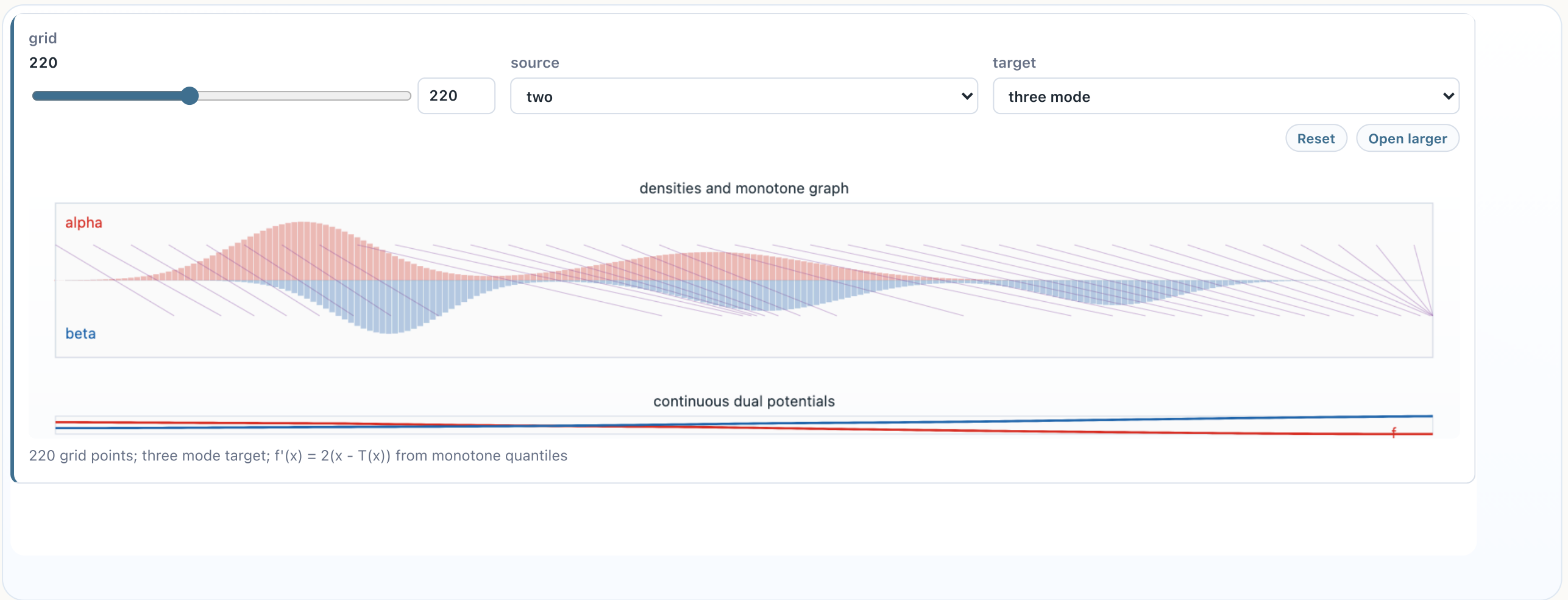
Here “Polish” means complete and separable after choosing a compatible metric. This is the infinite-dimensional analogue of the finite LP dual.

# Continuous potentials

For quadratic cost in one dimension, optimal potentials encode the monotone transport map through their derivatives.



# Live panel: continuous Kantorovich potentials



In one dimension, the contact condition can be read directly from the potential curves.

# c-transform

## c-transform.

For a function  $f$  on the source space  $X$ , define

$$f^c(y) = \inf_{x \in X} (c(x, y) - f(x)).$$

Similarly, for  $g$  on the target space  $Y$ ,

$$g^{\bar{c}}(x) = \inf_{y \in Y} (c(x, y) - g(y)).$$

The dual can be optimized by replacing one potential by the largest admissible companion.

# Reduced dual value

The  $\mathcal{C}$ -transform removes one dual variable:

$$\mathcal{L}_c(\alpha, \beta) = \sup_f \left\{ \int f d\alpha + \int f^c d\beta \right\}.$$

In the discrete case this is

$$\mathcal{L}_C(a, b) = \max_{f \in \mathbb{R}^n} \{ \langle f, a \rangle + \langle f^C, b \rangle \}, \quad f_j^C = \min_i (C_{i,j} - f_i).$$

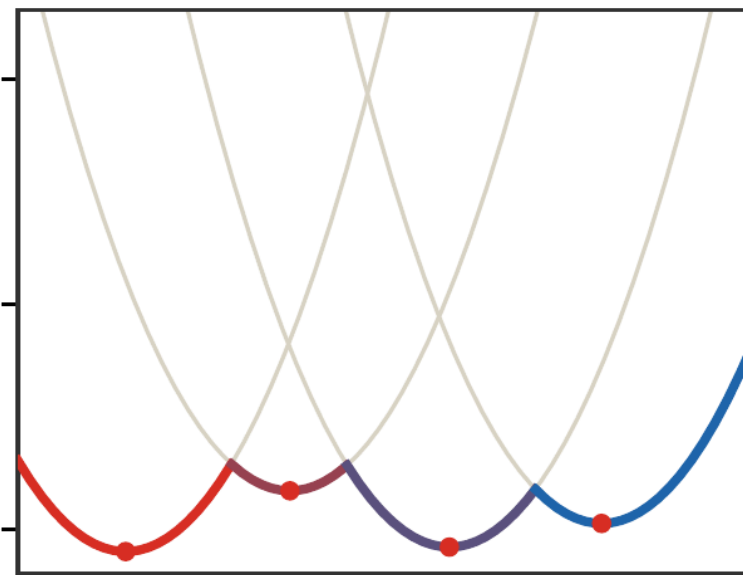
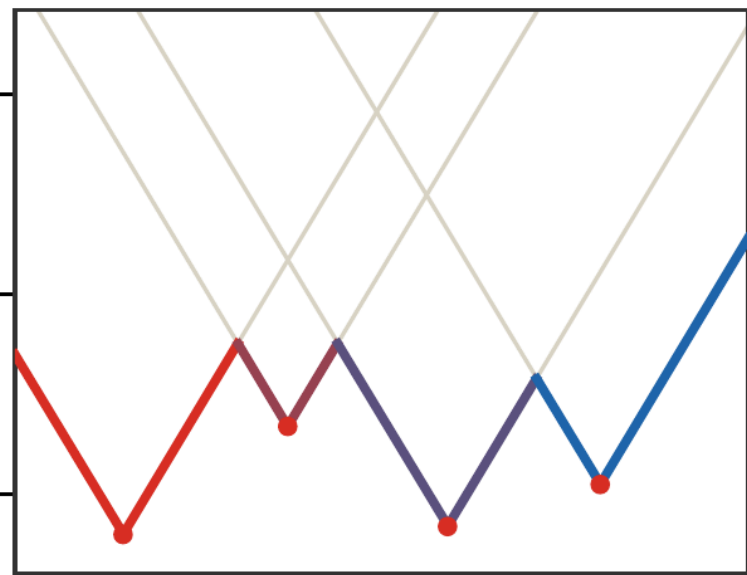
This is the formula behind alternating  $\mathcal{C}$ -transform updates.

## Reduction to one potential.

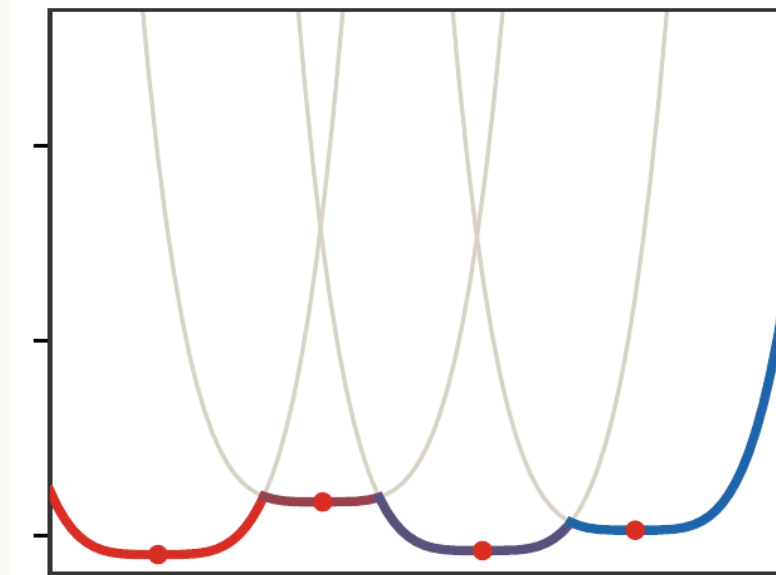
At optimum one can impose

$$g = f^c, \quad f = g^{\bar{c}}.$$

Thus dual potentials are fixed points of alternating c-transforms, up to non-uniqueness.



OT4ML - Duality and Semi-discrete OT



# Triple c-transform closure

The identity  $f^{ccc} = f^c$  explains why alternating c-transforms stabilize.

## Closure identity.

For the c-transform, one always has

$$f^{ccc} = f^c,$$

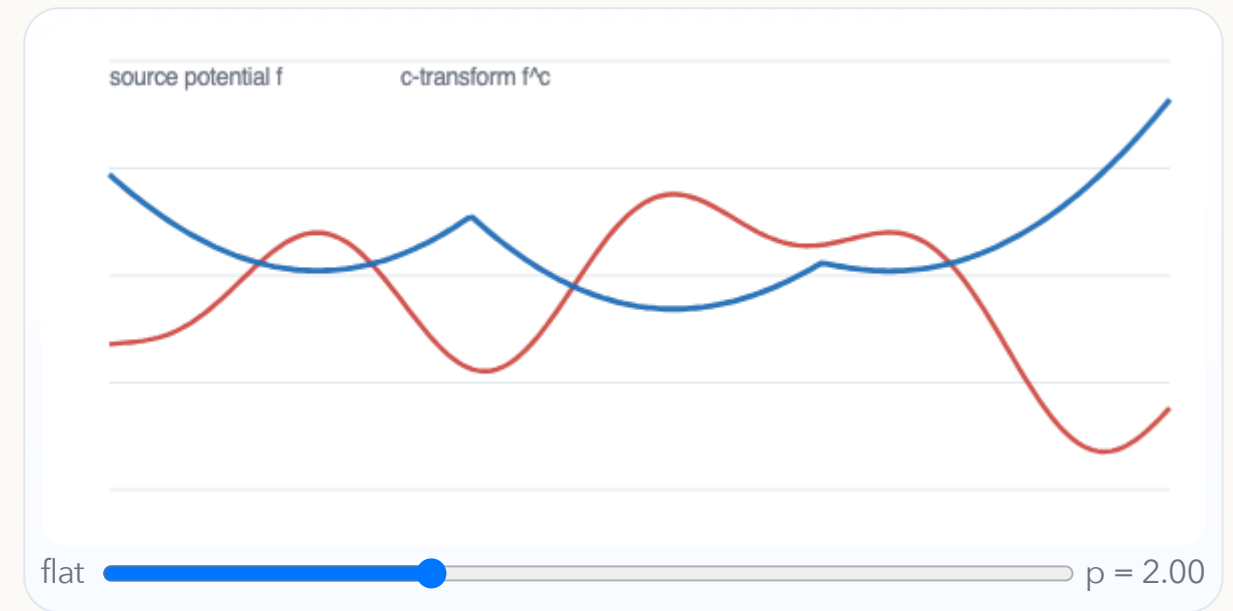
with the appropriate alternation between  $X$  and  $Y$ .

Proof idea: the inequality  $f^{cc} \geq f$  implies  $f^{ccc} \leq f^c$ , while applying the defining inequality of the c-transform in the reverse direction gives  $f^{ccc} \geq f^c$ .

# Interactive: changing the cost exponent

The  $c$ -transform closes a potential under the constraint  $f(x) + g(y) \leq c(x, y)$ . Changing the cost exponent changes the envelope selected by the dual constraint.

Here  $c_p(x, y) = |x - y|^p$  and the slider controls  $p$ .





# Alternating c-transforms

Alternating minimization in the dual repeatedly closes potentials by c-transforms:

## Alternating c-transform closure.

Input: initial source potential  $f$ .

Output: dual feasible potentials  $(f, g)$  after stabilization.

1. Set an initial source potential  $f$ .

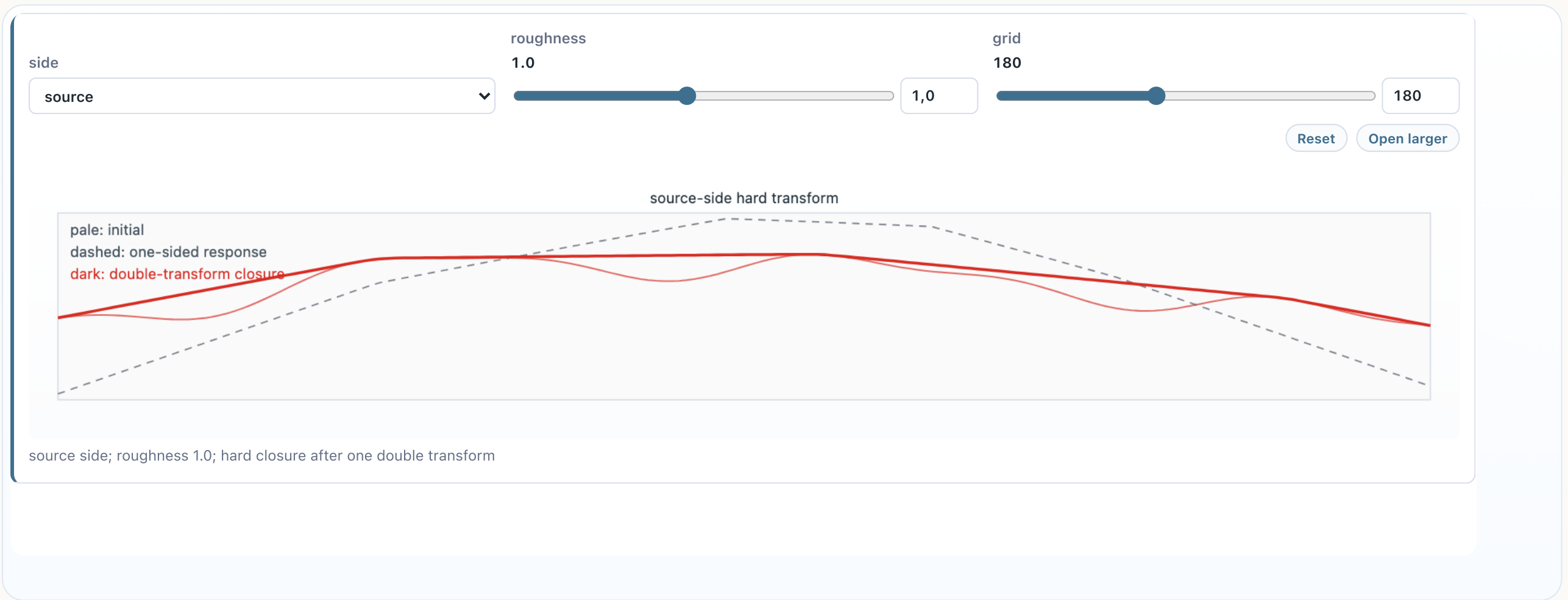
2. Repeat the closure updates

$$g \leftarrow f^c, \quad f \leftarrow g^{\bar{c}}.$$

3. Stop when the potentials are numerically stable.

Useful warning: this reveals the dual structure, but it is not by itself a fast generic OT solver.

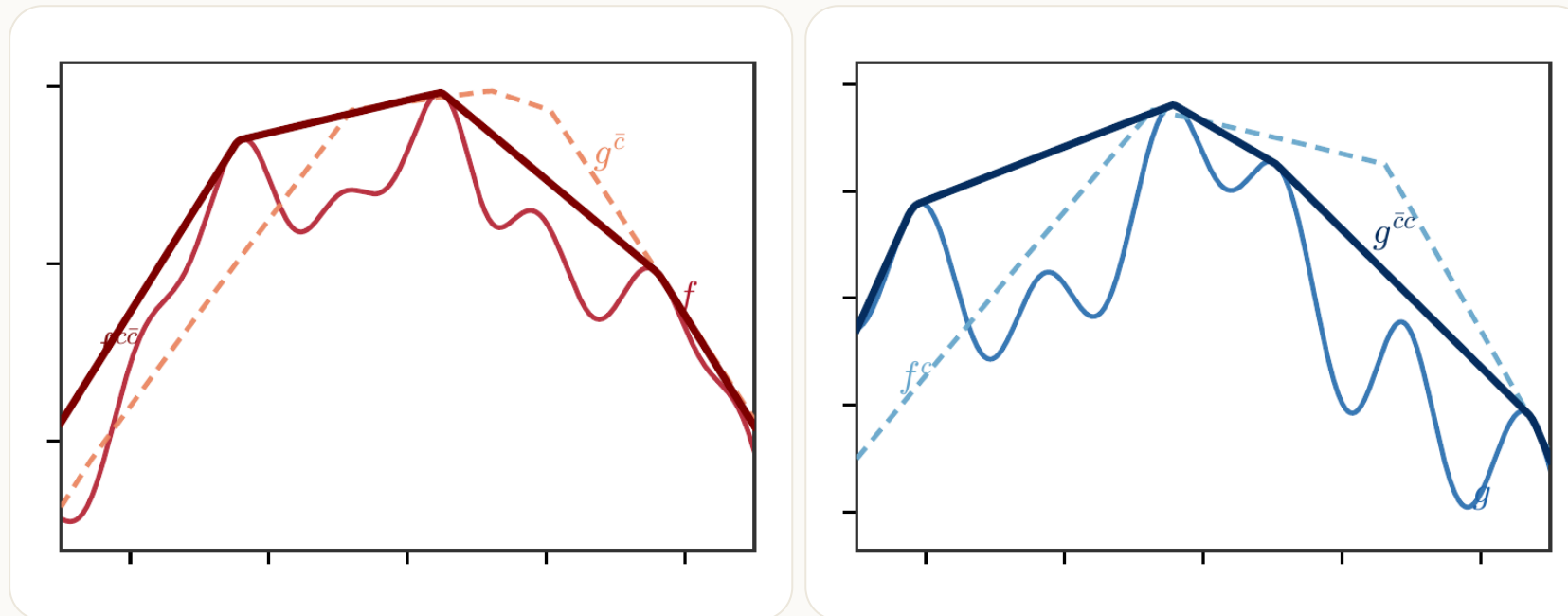
# Live panel: alternating c-transforms



Alternating envelopes implement the coordinate ascent geometry of the dual.

# c-transform closure geometry

The source and target closures can be viewed as envelopes of admissible potentials.



# Live panel: c-transform envelopes



The envelope picture explains why c-concave potentials are the stable objects.

# Existence of dual optimizers

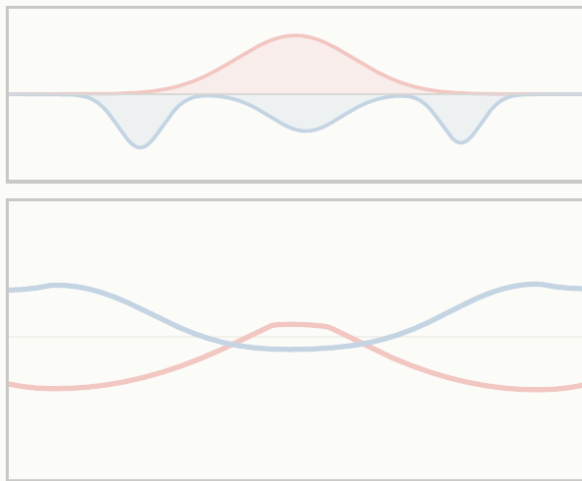
## **Lipschitz compactness principle.**

When the cost is Lipschitz on compact supports, optimal potentials can be chosen uniformly Lipschitz. Arzela-Ascoli compactness then yields an optimizer.

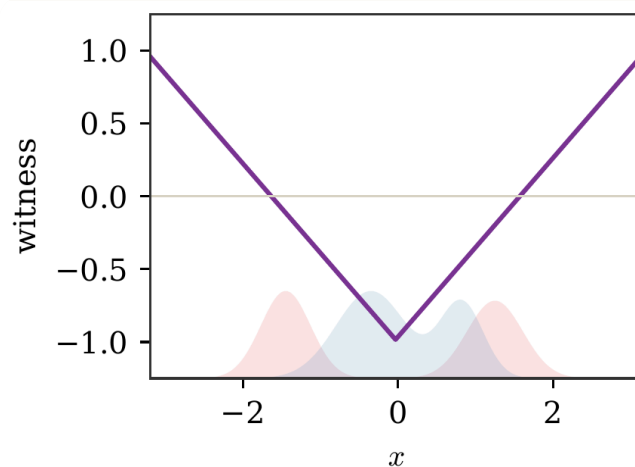
This proof idea is important because the dual problem is over functions, not finite-dimensional vectors.

# Roadmap: Wasserstein-1 and integral probability metrics

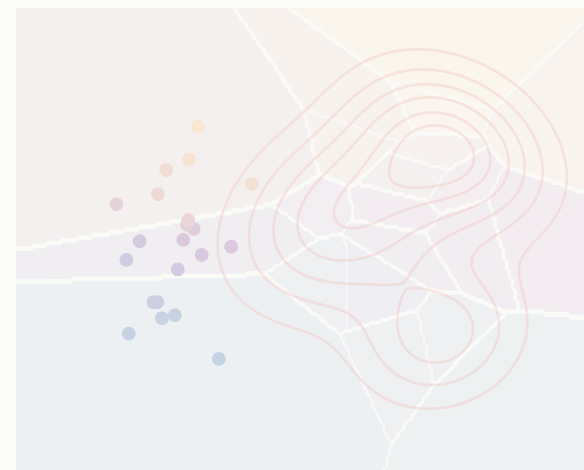
## 1. Duality



## 2. Wasserstein-1 and IPMs



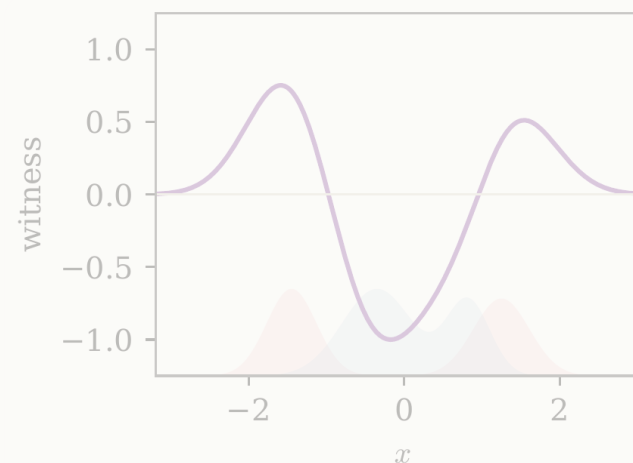
## 3. Semi-discrete OT



## 4. Quantization



## 5. Learning links



OT4ML - Duality and Semi-discrete OT

# Kantorovich-Rubinstein formula

For  $c(\textcolor{red}{x}, \textcolor{blue}{y}) = d(\textcolor{red}{x}, \textcolor{blue}{y})$ , the dual simplifies dramatically.

## Wasserstein-1 duality.

$$W_1(\textcolor{red}{\alpha}, \textcolor{blue}{\beta}) = \sup_{\text{Lip}(f) \leq 1} \int f \, d(\textcolor{red}{\alpha} - \textcolor{blue}{\beta}).$$

Here

$$\text{Lip}(f) = \sup_{\textcolor{red}{x} \neq \textcolor{blue}{y}} \frac{|f(\textcolor{red}{x}) - f(\textcolor{blue}{y})|}{d(\textcolor{red}{x}, \textcolor{blue}{y})}.$$

The witness  $f$  is a 1-Lipschitz discriminator between the two measures.

# Wasserstein-1 as a norm

## Norm viewpoint.

On signed measures with zero total mass,

$$\|\alpha - \beta\|_{W_1} := W_1(\alpha, \beta)$$

is a norm whenever the ground metric is fixed.

This places  $W_1$  among dual norms defined by classes of test functions.



# Integral probability metrics

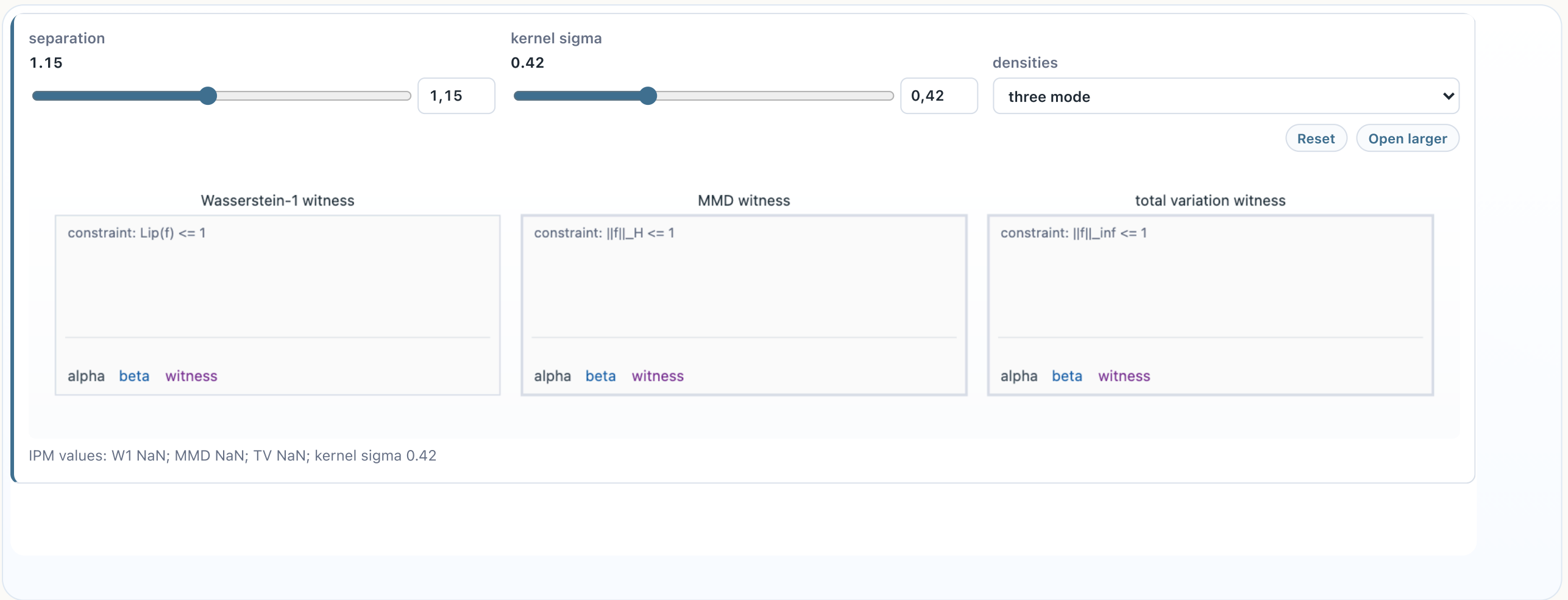
## Integral probability metric.

Given a symmetric class of test functions  $\mathcal{F}$ ,

$$D_{\mathcal{F}}(\alpha, \beta) = \sup_{f \in \mathcal{F}} \int f d(\alpha - \beta).$$

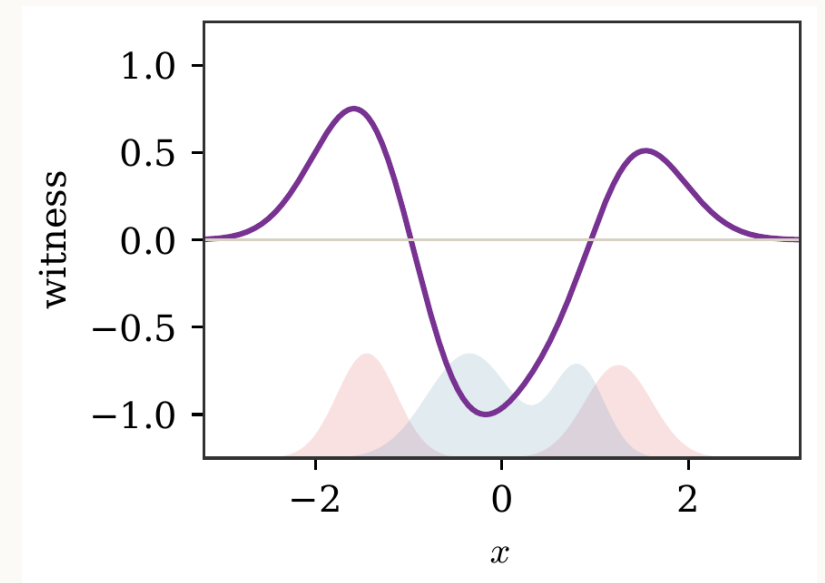
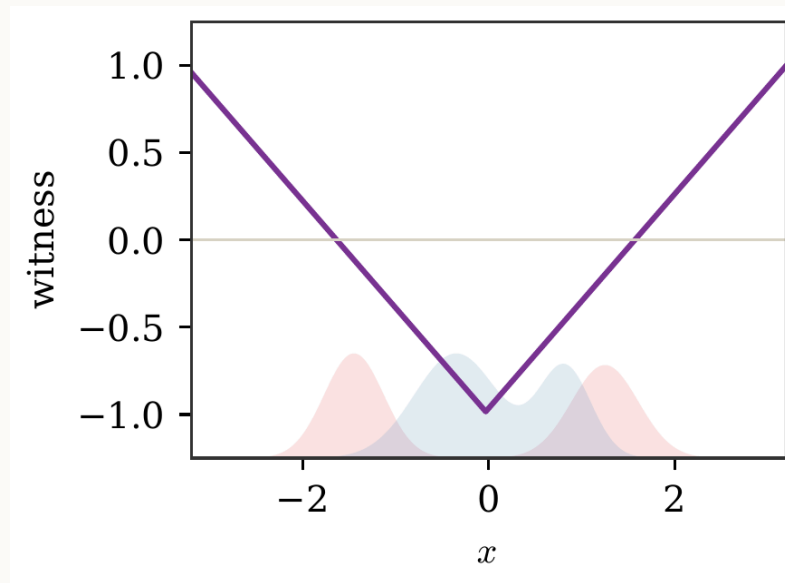
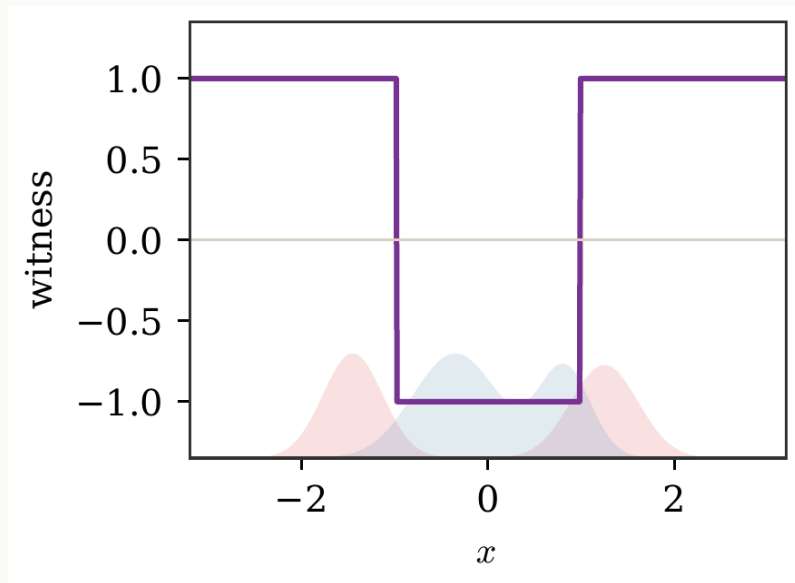
Examples include total variation, Wasserstein-1, bounded-Lipschitz distances, and MMDs.

# Live panel: integral probability metrics



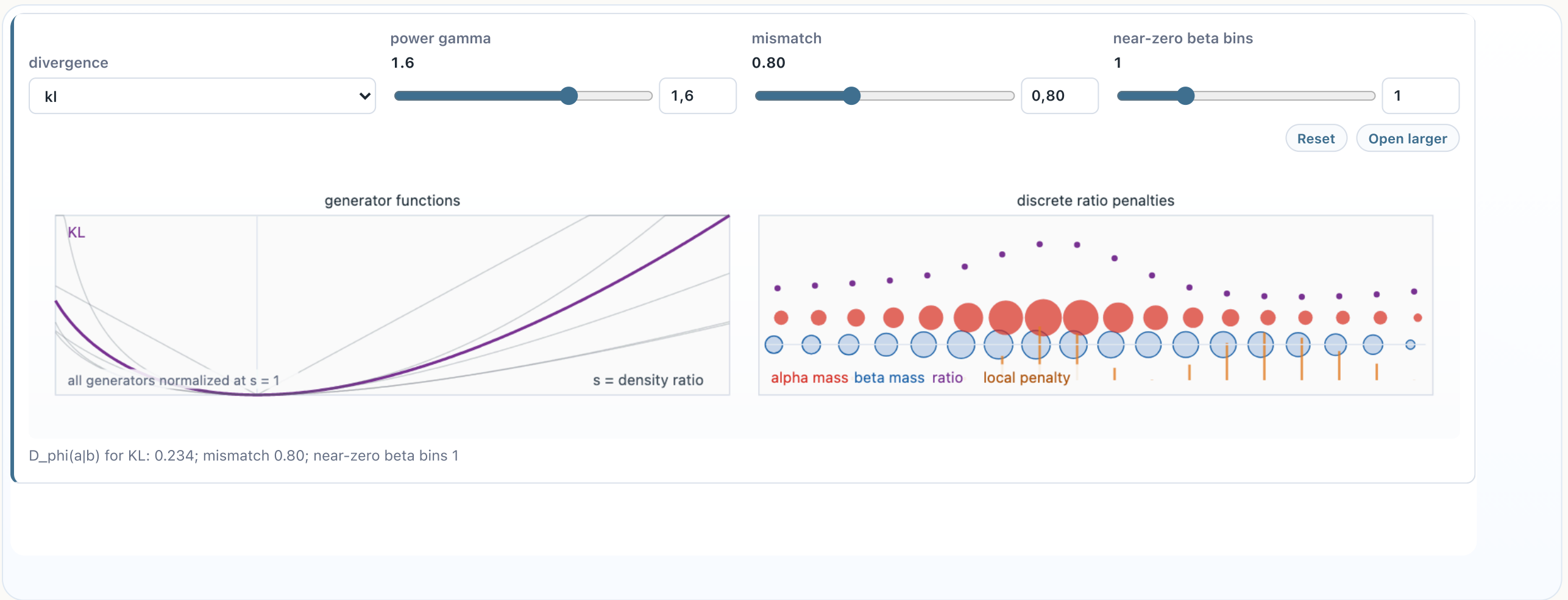
Changing the witness class changes the geometry of the discrepancy.

# Dual witnesses compared



Different function classes produce different witnesses and therefore different notions of discrepancy.

# Live panel: witness constraints



The same signed mass can be tested by very different admissible witnesses.

# Wasserstein-1 on Euclidean domains

For smooth  $f$  on Euclidean domains,  $\text{Lip}(f) \leq 1$  is equivalent to

$$\|\nabla f(x)\| \leq 1.$$

This connects  $W_1$  to Beckmann's continuous flux formulation:

$$W_1(\alpha, \beta) = \min_m \int \|\mathfrak{m}(x)\| dx \quad \text{s.t.} \quad \text{div } \mathfrak{m} = \alpha - \beta.$$

# W1 on surfaces and subdomains

The same Beckmann idea extends beyond flat Euclidean grids by replacing gradients and divergence with their geometric counterparts.

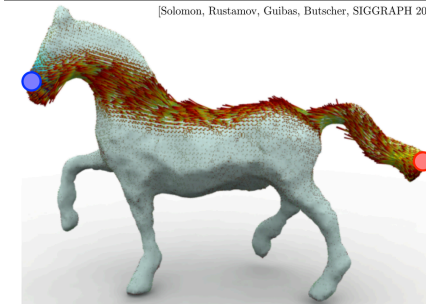
## Geometric message.

On meshes, surfaces, or subdomains,  $W_1$  can be computed as a flow constrained by the geometry. Graph discretizations are the simplest finite-dimensional version of this idea.

This is the setting of surface and subdomain  $W_1$  solvers used in geometry processing, including the Solomon-Rustamov-Guibas-Butscher line of work. Graph flows are the simplest discrete proxy for this geometric formulation.

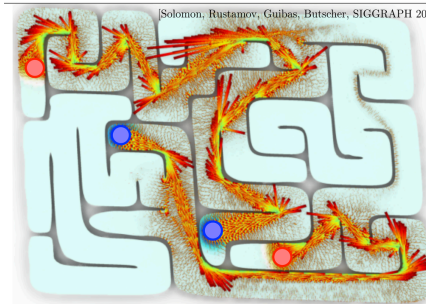
W1 On Surfaces

[Solomon, Rustamov, Guibas, Butscher, SIGGRAPH 2014]



W1 On Sub-domains

[Solomon, Rustamov, Guibas, Butscher, SIGGRAPH 2014]



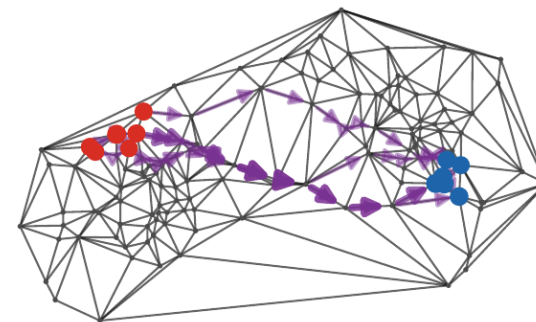
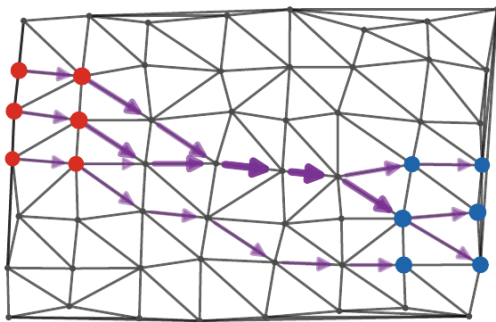
# Wasserstein-1 on graphs

On a graph, the Beckmann formulation becomes a sparse network-flow problem over edges:

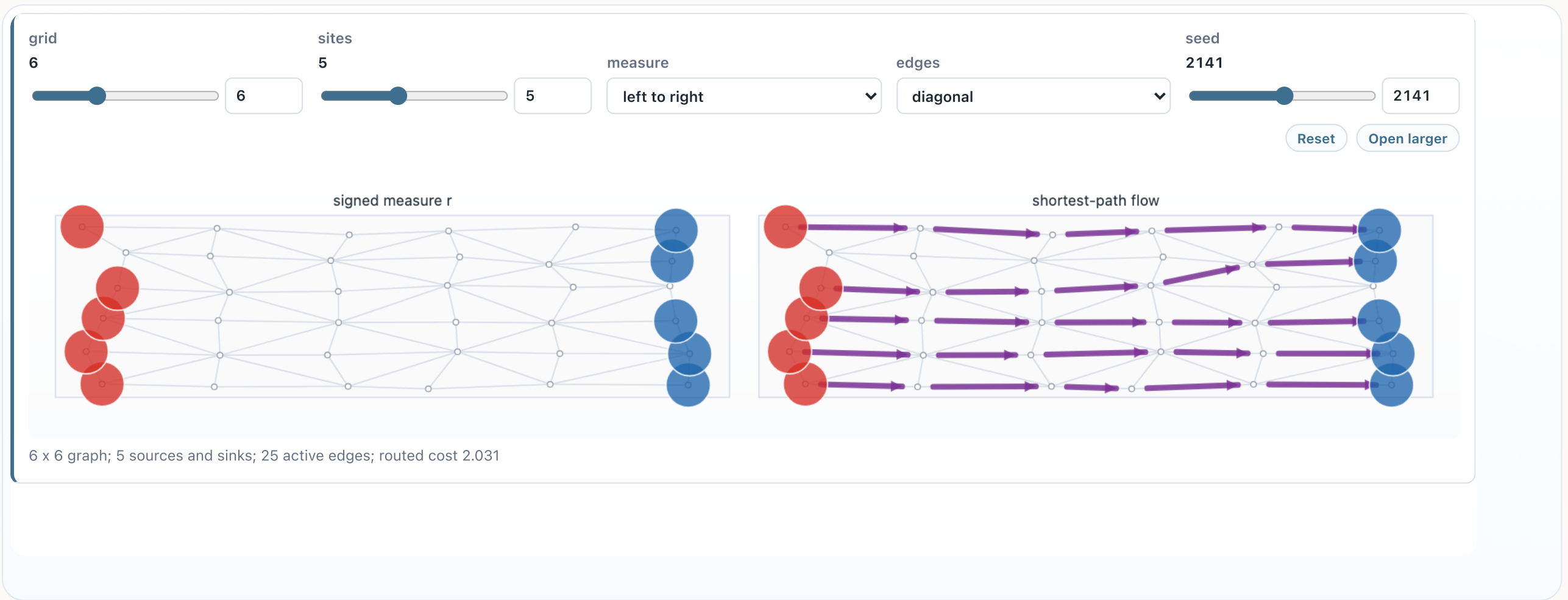
$$\min_q \sum_e w_e |q_e| \quad \text{s.t.} \quad Dq = a - b.$$

Here  $D$  is the signed vertex-edge incidence matrix and  $q_e$  is the oriented mass flux on edge  $e$ .

This can be much smaller than a full pairwise Kantorovich LP.



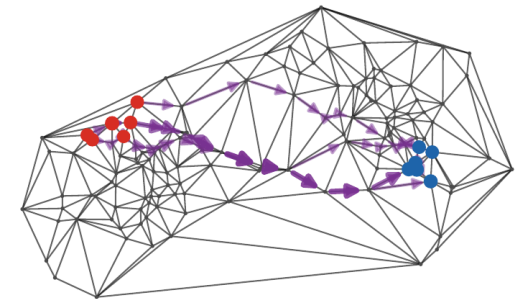
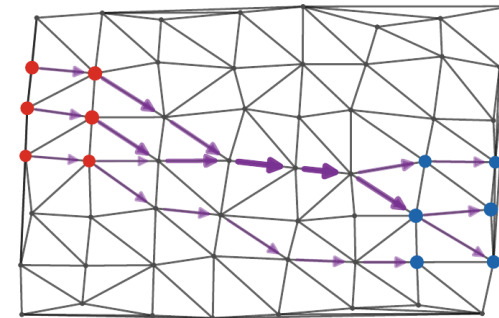
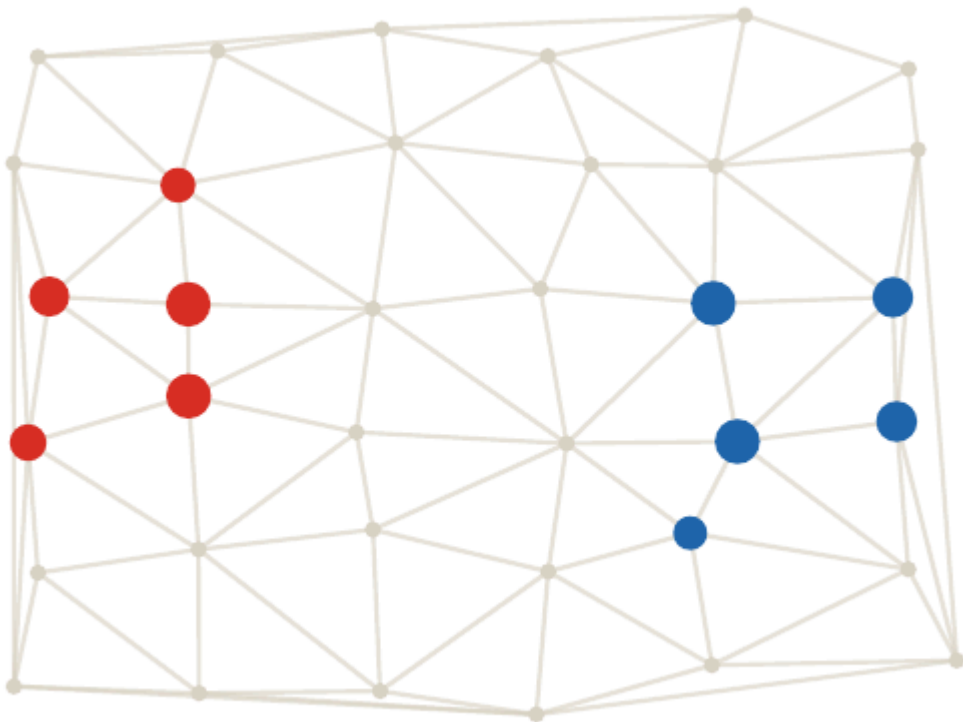
# Live panel: graph Beckmann transport



The graph formulation turns W1 into a network flow problem.



# Graph flow visualization



The edge flow is the sparse object that transports positive mass to negative mass through the graph geometry.

# WGAN viewpoint

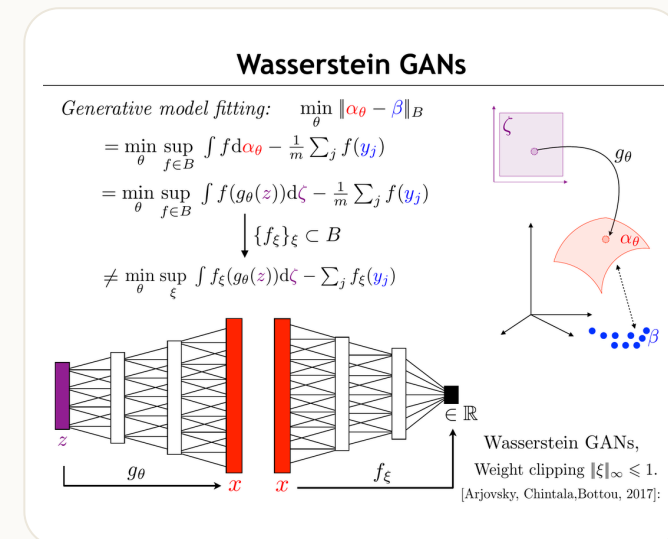
## Example: Wasserstein GAN.

A generator  $g_\theta$  can be trained by minimizing

$$\min_{\theta} W_1((g_\theta)_\# \zeta, \beta) = \min_{\theta} \sup_{\text{Lip}(f) \leq 1} \mathbb{E} f(g_\theta(Z)) - \mathbb{E} f(Y).$$

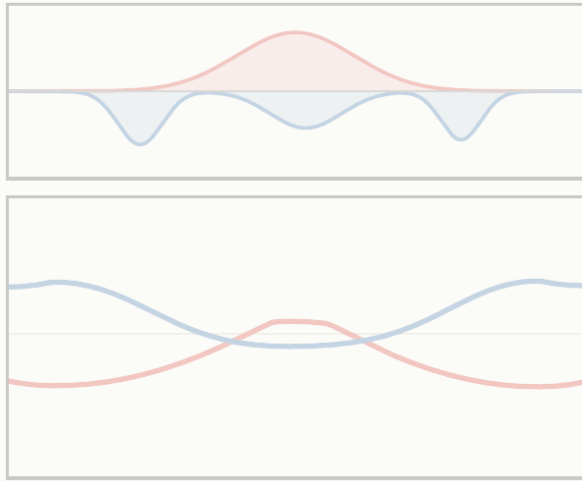
The discriminator is a parametrized approximation of a 1-Lipschitz Kantorovich potential.

This is the machine-learning face of Kantorovich-Rubinstein duality.

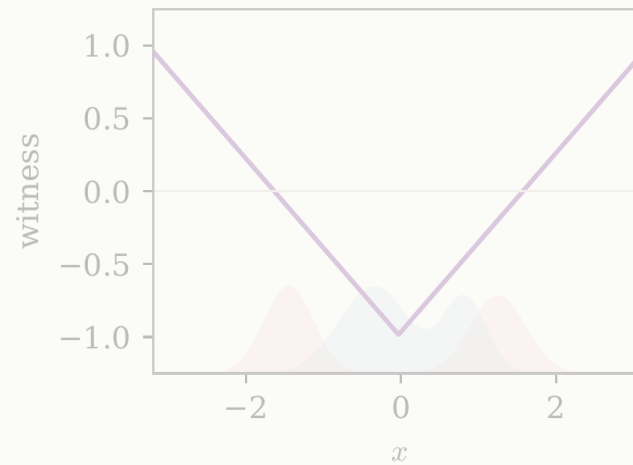


# Roadmap: semi-discrete OT

## 1. Duality



## 2. Wasserstein-1 and IPMs



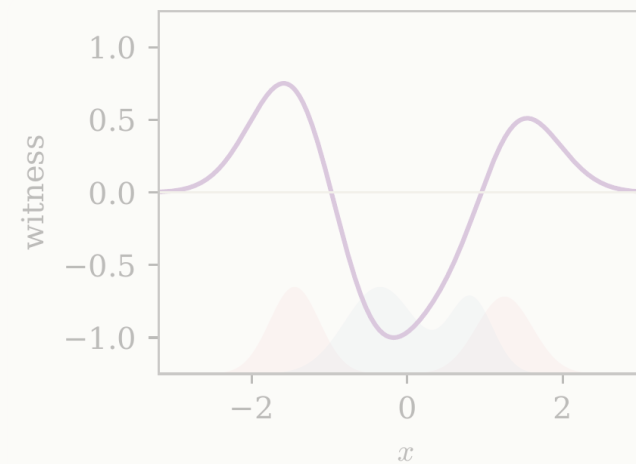
## 3. Semi-discrete OT



## 4. Quantization



## 5. Learning links



# Semi-discrete setting

## Semi-discrete OT.

Let  $\alpha$  be a continuous probability measure on  $\mathbb{R}^d$  and

$$\beta = \sum_{j=1}^m b_j \delta_{y_j}$$

be discrete. For quadratic cost, one seeks a partition of the **source** domain whose cells have masses  $b_j$ .

The unknown is not a full coupling matrix but a weighted spatial partition.

# Laguerre cells

## Laguerre cell.

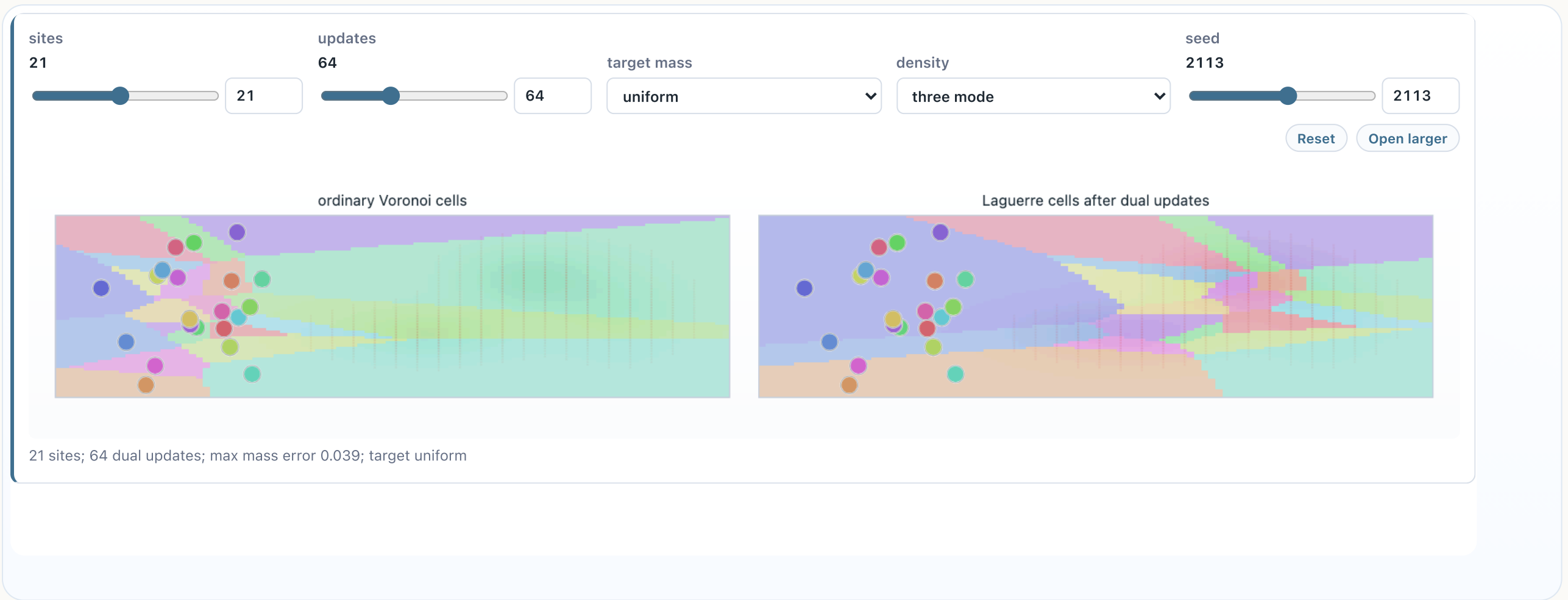
For target sites  $y_j$  and weights  $g_j$ , the cell associated with  $j$  is

$$L_j(g) = \{x : \|x - y_j\|^2 - g_j \leq \|x - y_k\|^2 - g_k \quad \forall k\}.$$

The semi-discrete **map** sends every  $x \in L_j(g)$  to  $y_j$ . Optimal weights are characterized by the mass constraints

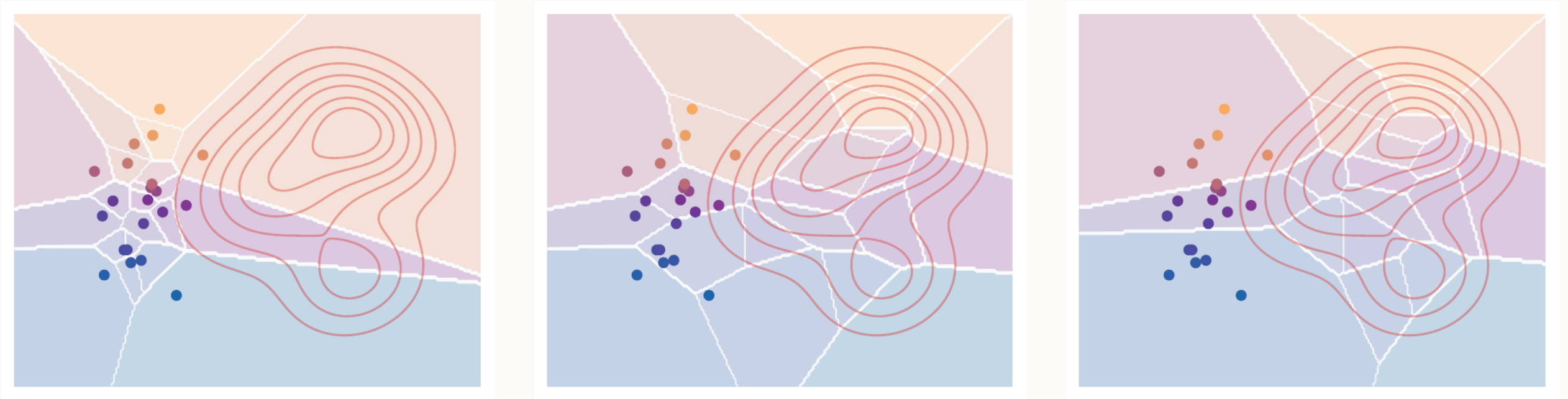
$$\alpha(L_j(g)) = b_j, \quad j = 1, \dots, m.$$

# Live panel: Laguerre cells and dual weights



Changing a target weight geometrically expands or shrinks its cell.

# Power diagrams



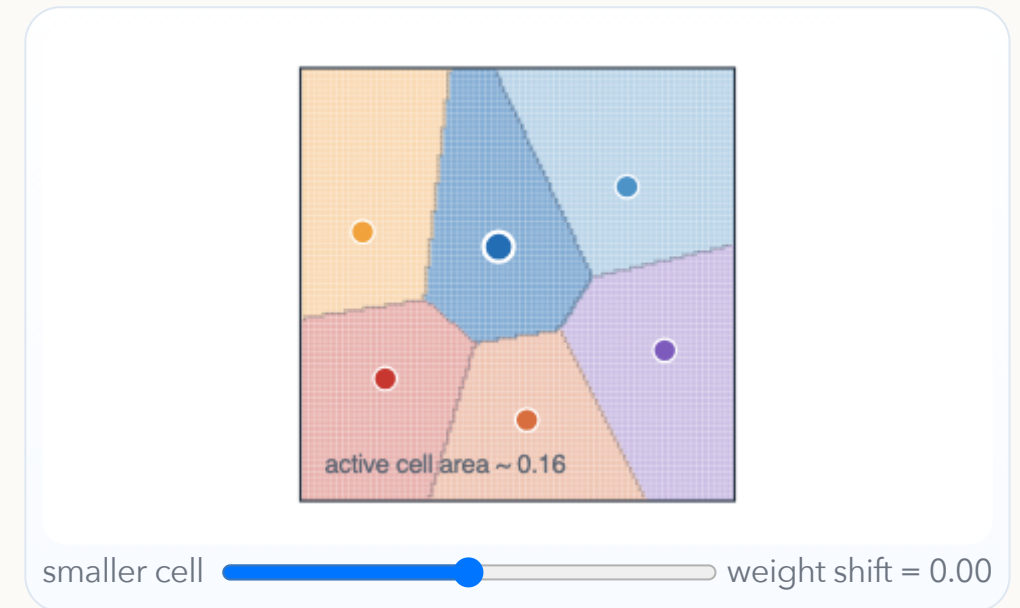
Voronoi cells become Laguerre cells when weights are introduced. The weights tune cell masses.

# Interactive: one weight changes one cell

Changing  $g_j$  shifts the additive power distance

$$\|\mathbf{x} - \mathbf{y}_j\|^2 - g_j.$$

Increasing  $g_j$  makes site  $j$  more attractive and expands its Laguerre cell.





# Semi-discrete dual

For quadratic cost, the dual can be written as a concave maximization over weights:

$$\max_{g \in \mathbb{R}^m} \int \min_j (\|x - y_j\|^2 - g_j) d\alpha(x) + \sum_{j=1}^m b_j g_j.$$

## Gradient.

The gradient component is

$$\frac{\partial}{\partial g_j} = b_j - \alpha(L_j(g)).$$

Thus dual ascent balances Laguerre cell masses.

# Hessian and Newton structure

The reference semi-discrete solver uses the fact that the dual has an explicit sparse Hessian. If  $\alpha = \rho(x) dx$  and cells  $L_i(g)$  and  $L_j(g)$  share a facet

$$F_{i,j} = L_i(g) \cap L_j(g),$$

then for  $i \neq j$ ,

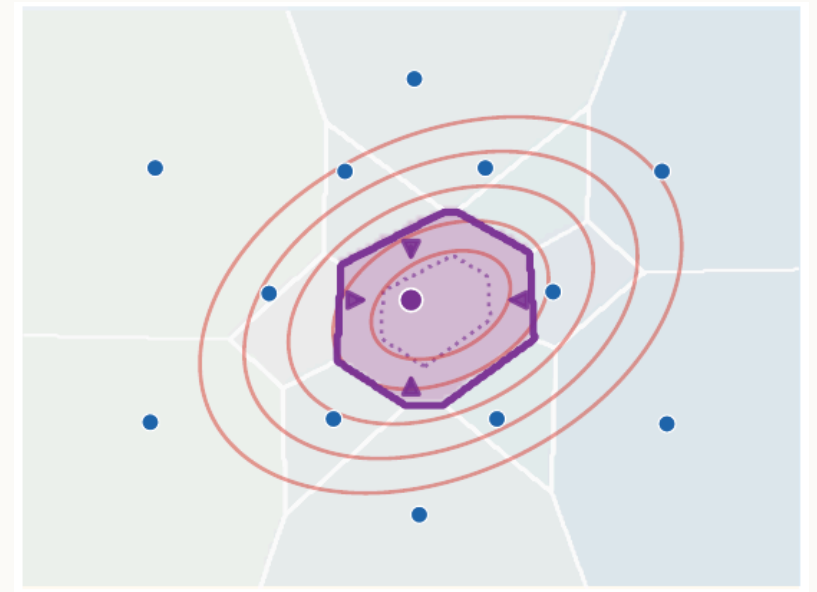
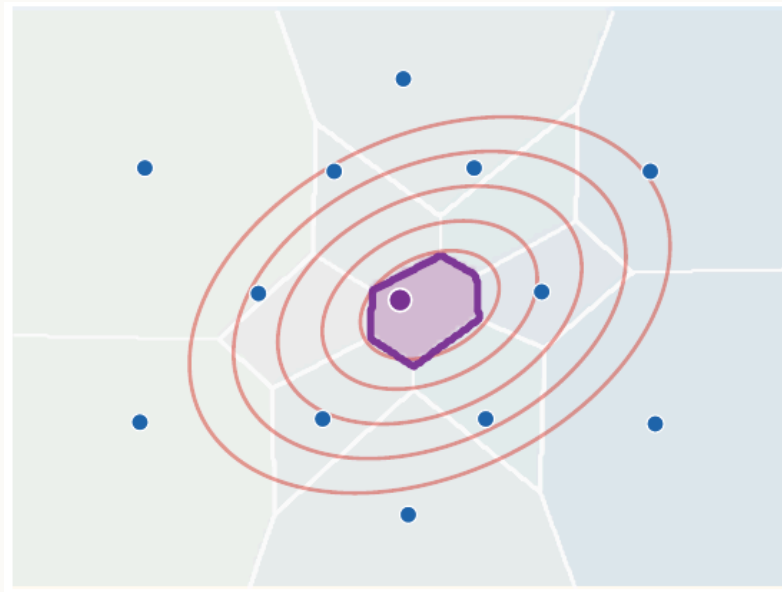
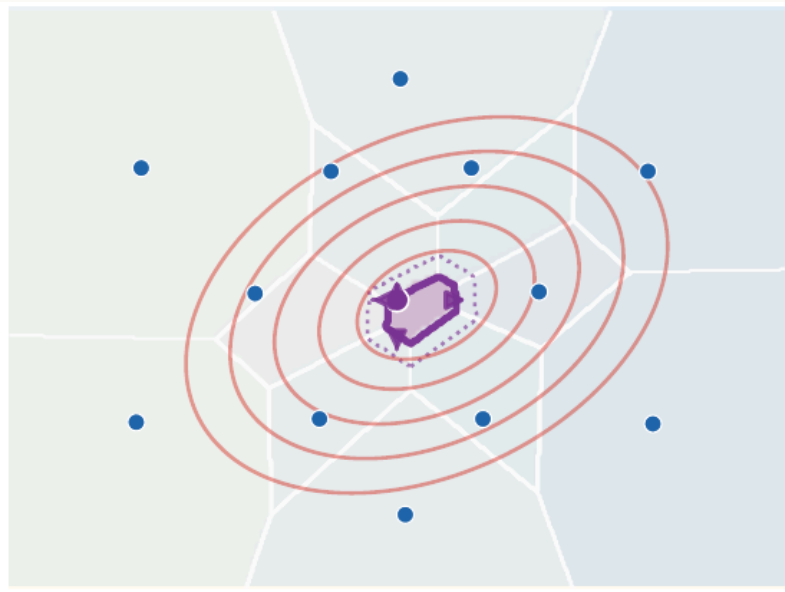
$$\frac{\partial \alpha(L_i(g))}{\partial g_j} = -\frac{1}{2\|y_i - y_j\|} \int_{F_{i,j}} \rho(x) d\mathcal{H}^{d-1}(x),$$

and the diagonal is chosen so rows sum to zero:

$$\frac{\partial \alpha(L_i(g))}{\partial g_i} = -\sum_{j \neq i} \frac{\partial \alpha(L_i(g))}{\partial g_j}.$$

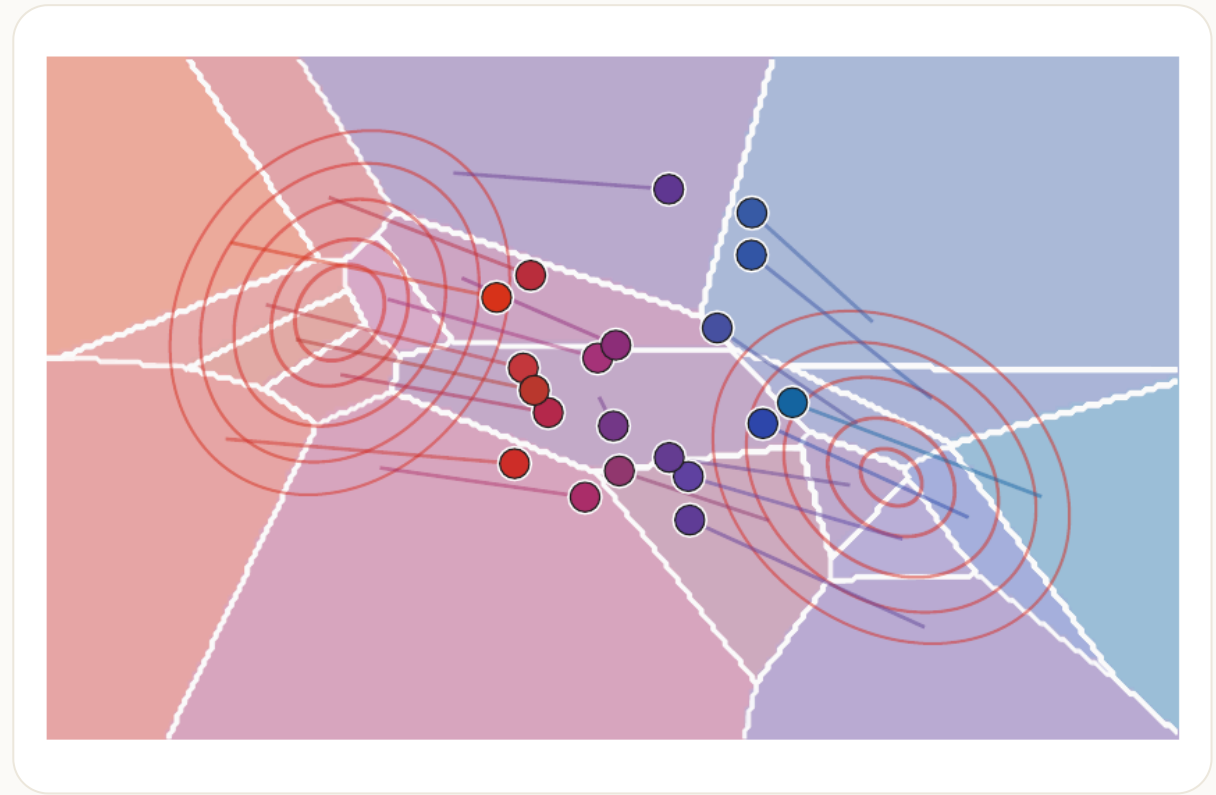
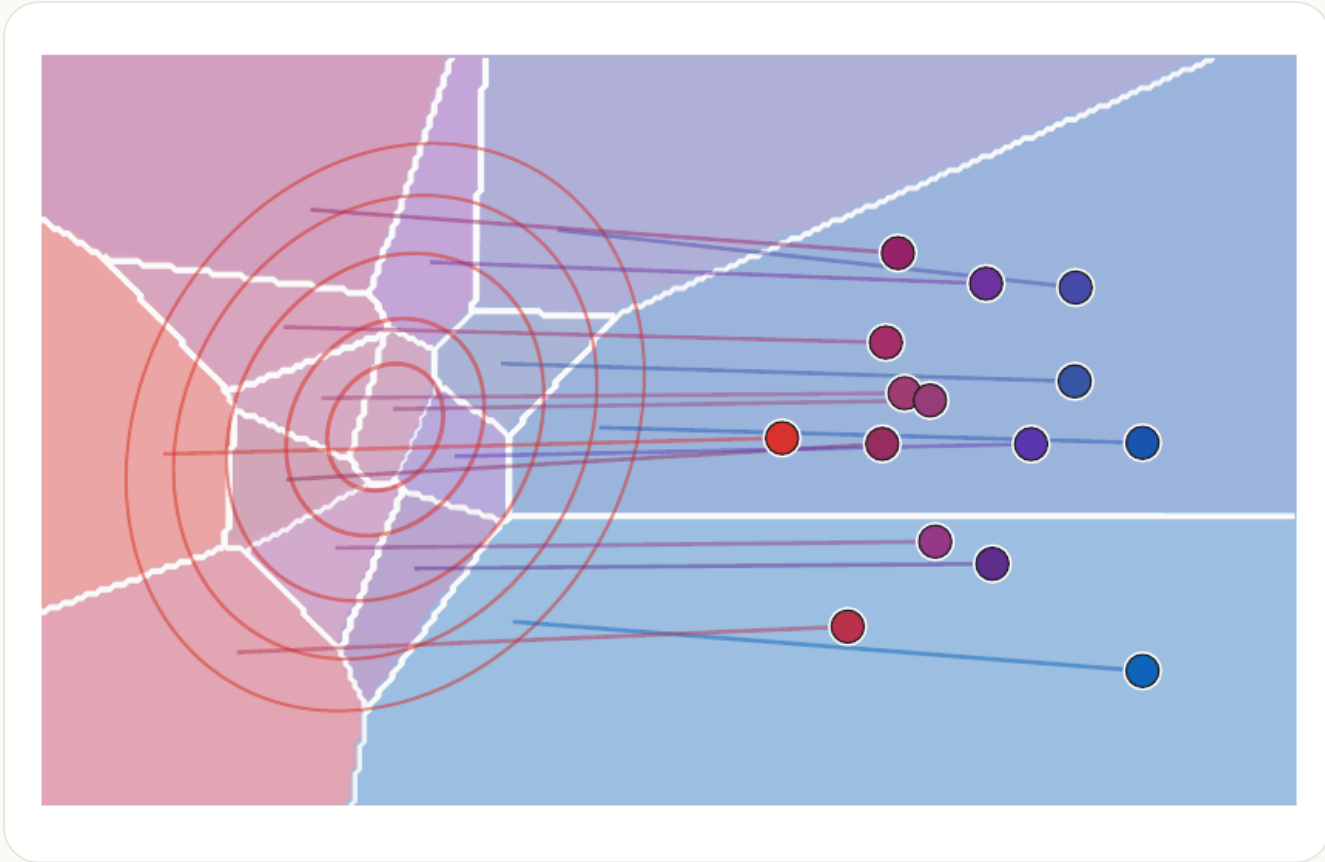
Thus the Hessian of the concave dual has positive off-diagonal entries and negative diagonal row sums. Newton or quasi-Newton methods exploit the adjacency graph of the Laguerre diagram rather than a dense coupling matrix.

# Weight descent as mass balancing



The algorithm updates the weights until every cell has the prescribed mass.

# Semi-discrete examples



Coloring makes the spatial map visible: each source cell is sent to one target atom.

# Practical optimization

## Semi-discrete dual ascent.

Input: source density  $\alpha$ , target sites  $y_j$ , target masses  $b_j$ .

Initialize weights  $g$ .

Repeat:

$$L_j(g) = \{x : \|x - y_j\|^2 - g_j \leq \|x - y_k\|^2 - g_k\},$$

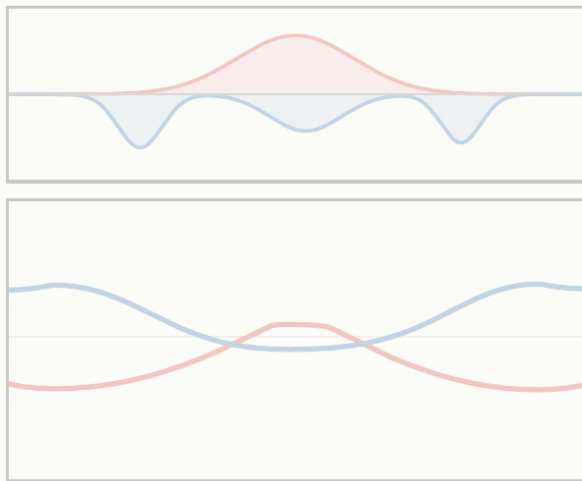
$$g_j \leftarrow g_j + \tau(b_j - \alpha(L_j(g))).$$

Output: the Laguerre partition and map  $T(x) = y_j$  for  $x \in L_j(g)$ .

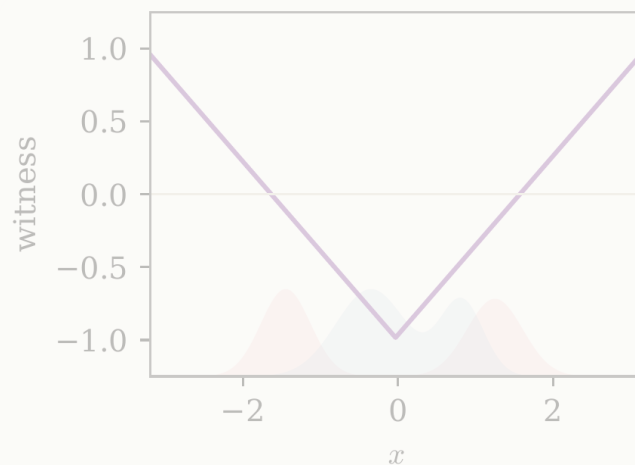
Newton and quasi-Newton methods, especially L-BFGS variants, are typically used in high-quality implementations because the semi-discrete dual is a smooth concave objective away from cell degeneracies.

# Roadmap: optimal quantization

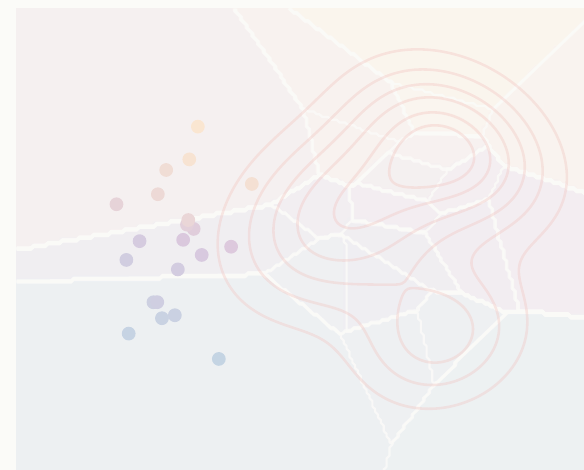
1. Duality



2. Wasserstein-1 and IPMs



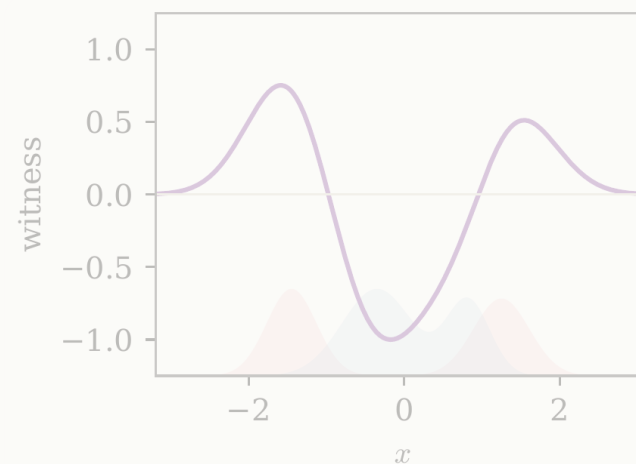
3. Semi-discrete OT



4. Quantization



5. Learning links



OT4ML - Duality and Semi-discrete OT

# Quantization problem

## Optimal quantization.

For a source measure  $\alpha$  and  $m$  target sites  $Y = (y_1, \dots, y_m)$ ,

$$Q_m(\alpha) = \min_{Y, b \in \Sigma_m} W_2^2 \left( \alpha, \sum_{j=1}^m b_j \delta_{y_j} \right).$$

For fixed sites  $Y$ , this equals

$$\int \min_j \|x - y_j\|^2 d\alpha(x)$$

when the masses are free.

The goal is to approximate a continuous measure by a small discrete measure.



# Fixed versus free masses

If the masses  $b_j$  are fixed, semi-discrete OT gives the optimal cells for fixed sites. If the masses are free, the optimal cells are Voronoi cells.

## Centroid condition.

At an unconstrained local optimum for quadratic cost,

$$y_j = \frac{\int_{V_j(Y)} x d\alpha(x)}{\alpha(V_j(Y))},$$

where  $V_j(Y)$  is the Voronoi cell of  $y_j$ .

# Quantization stationarity

For fixed sites  $Y$  and free masses, the semi-discrete dual collapses to the closest-site formula

$$\min_{b \in \Sigma_m} \mathcal{L}_c \left( \alpha, \sum_j b_j \delta_{y_j} \right) = \int \min_j c(x, y_j) d\alpha(x).$$

If the cells  $V_j(Y) = \{x : c(x, y_j) \leq c(x, y_k) \ \forall k\}$  are stable and  $c$  is differentiable in  $y$ , a local minimizer satisfies

$$\int_{V_j(Y)} \nabla_y c(x, y_j) d\alpha(x) = 0.$$

For  $c(x, y) = \|x - y\|^2$ , this is exactly the centroid condition.

# Lloyd algorithm

## Lloyd iteration.

Input: source measure  $\alpha$ , initial sites  $Y$ .

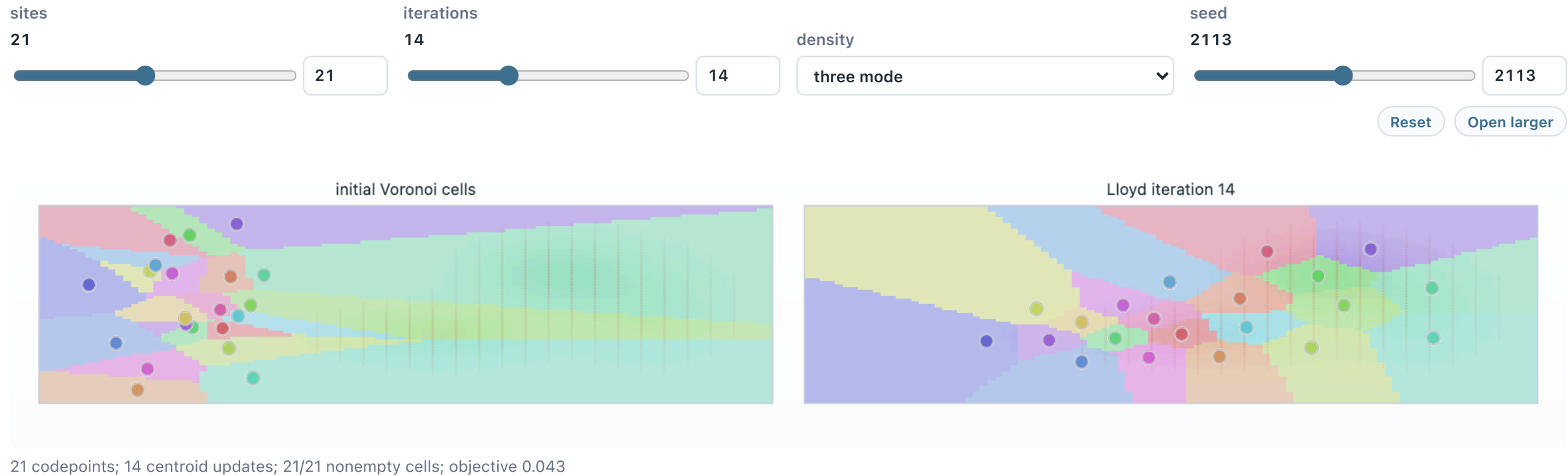
Repeat:

1. Build Voronoi cells  $V_j(Y)$ .
2. Replace each site by its cell centroid,

$$y_j \leftarrow \frac{\int_{V_j(Y)} x d\alpha(x)}{\alpha(V_j(Y))}.$$

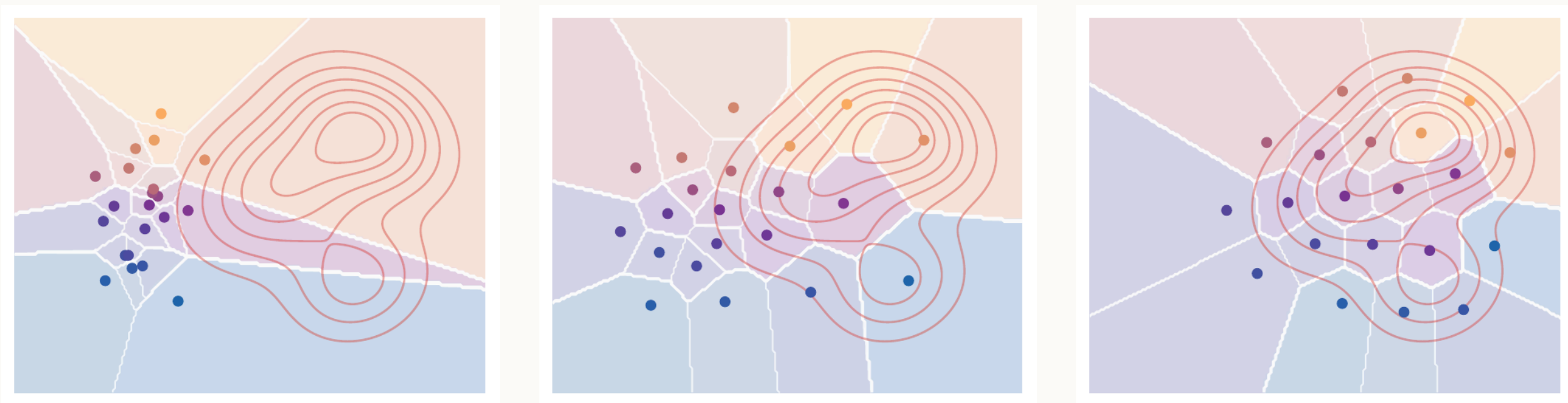
Output: a centroidal Voronoi tessellation.

# Live panel: Lloyd iterations



The iterations alternate cell assignment and centroid update.

# Lloyd iterations



Lloyd descent is simple and robust, but it is nonconvex and can converge to local minima.

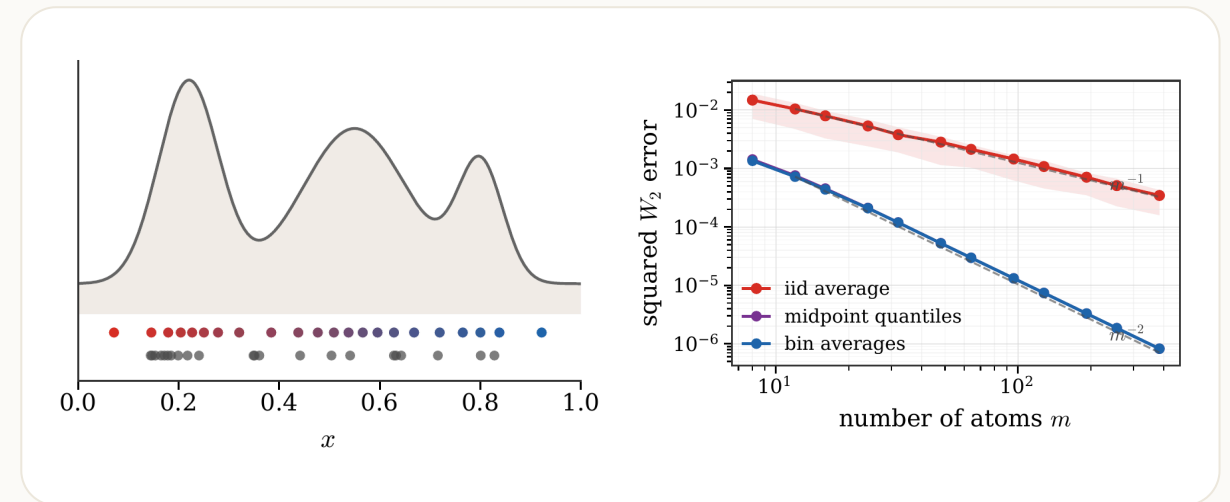
# Quantization on the line

In one dimension, quantiles give explicit optimal placements for equal masses. This is much better than random sampling for  $W_2$  approximation.

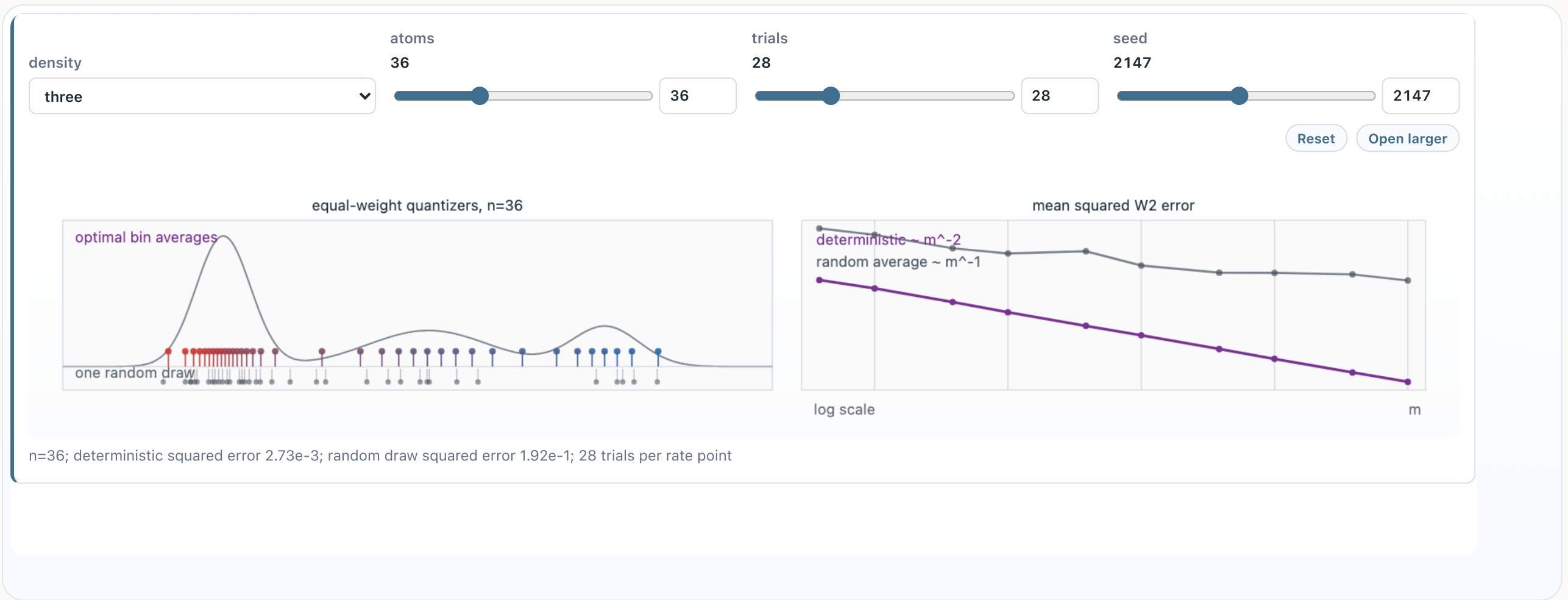
For  $\alpha$  with quantile  $F_\alpha^{-1}$ ,

$$y_j = F_\alpha^{-1}\left(\frac{j - \frac{1}{2}}{m}\right)$$

is the canonical equal-mass placement.



# Live panel: quantization rates

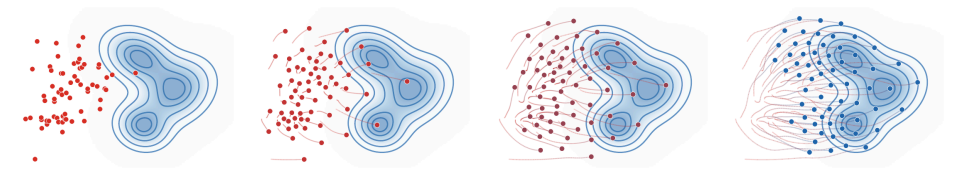


Optimal quantiles and random samples have visibly different convergence behavior.

# Relaxed Lloyd flow

A slowed-down Lloyd update can be viewed as a continuous-time descent on the quantization energy in configuration space. For small relaxation parameter  $\eta$ ,

$$y_j^{k+1} = (1 - \eta)y_j^k + \eta \text{centroid}(V_j(Y^k)).$$

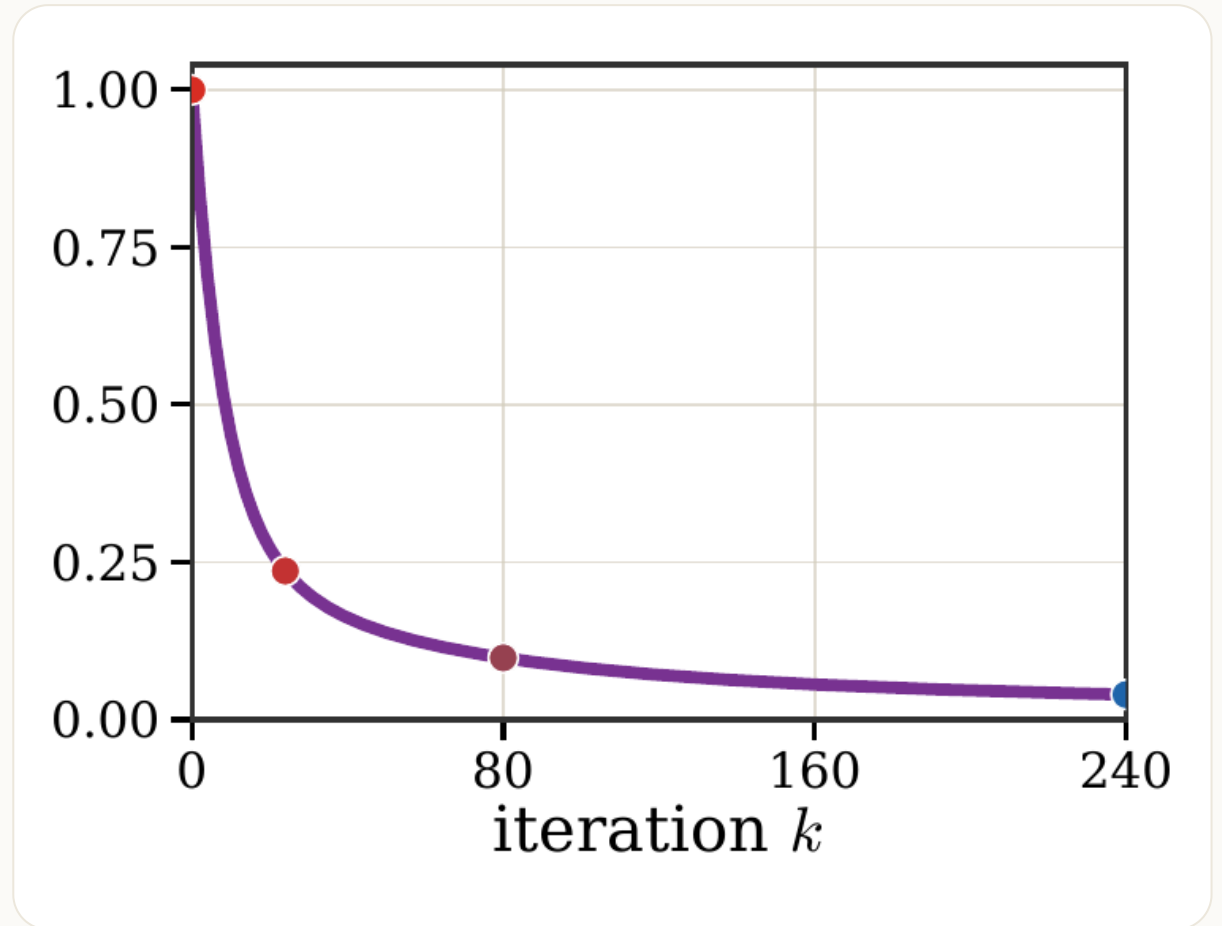


Sites move toward a quantized representation of the target density.



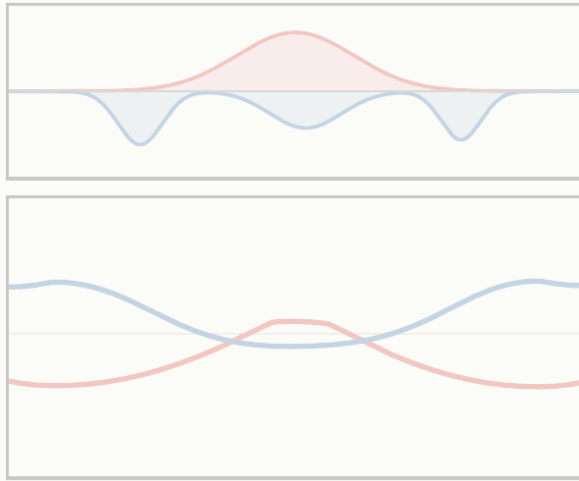
# Energy decay

The energy need not be globally convex, but standard Lloyd updates decrease it at each alternating minimization step.

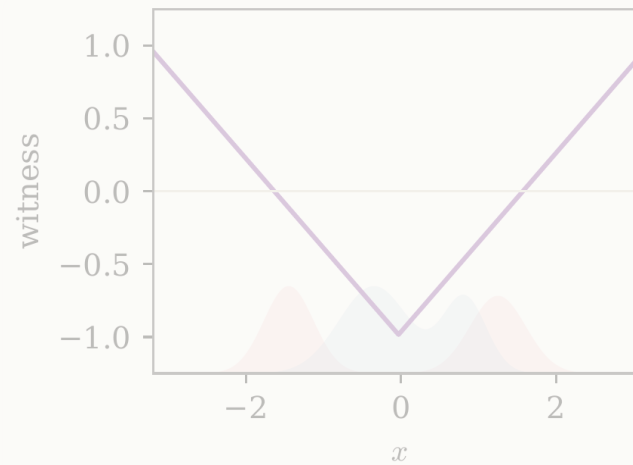


# Roadmap: links to learning

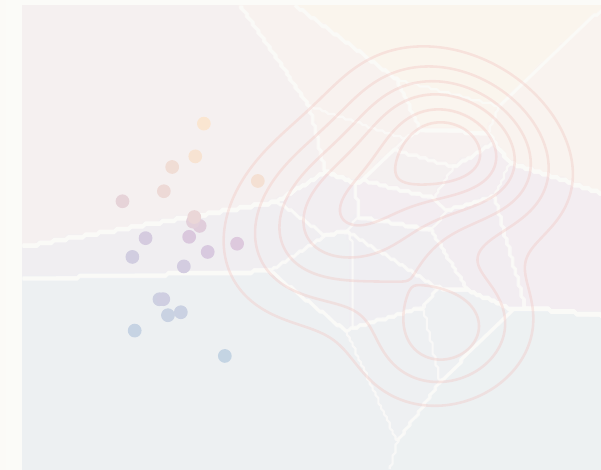
## 1. Duality



## 2. Wasserstein-1 and IPMs



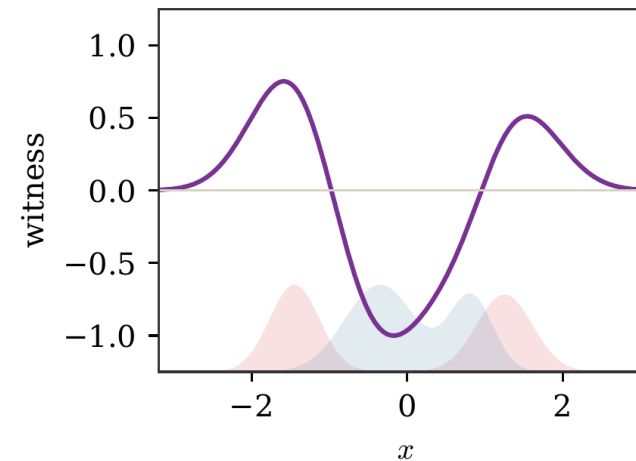
## 3. Semi-discrete OT



## 4. Quantization



## 5. Learning links



# Dual potentials as discriminators

The  $W_1$  dual is the mathematical core of Wasserstein GANs, while other IPMs replace the Lipschitz ball by another function class.

**TV**

bounded witnesses

**Wasserstein-1**

Lipschitz witnesses

**MMD**

RKHS witnesses

# Semi-discrete maps in applications

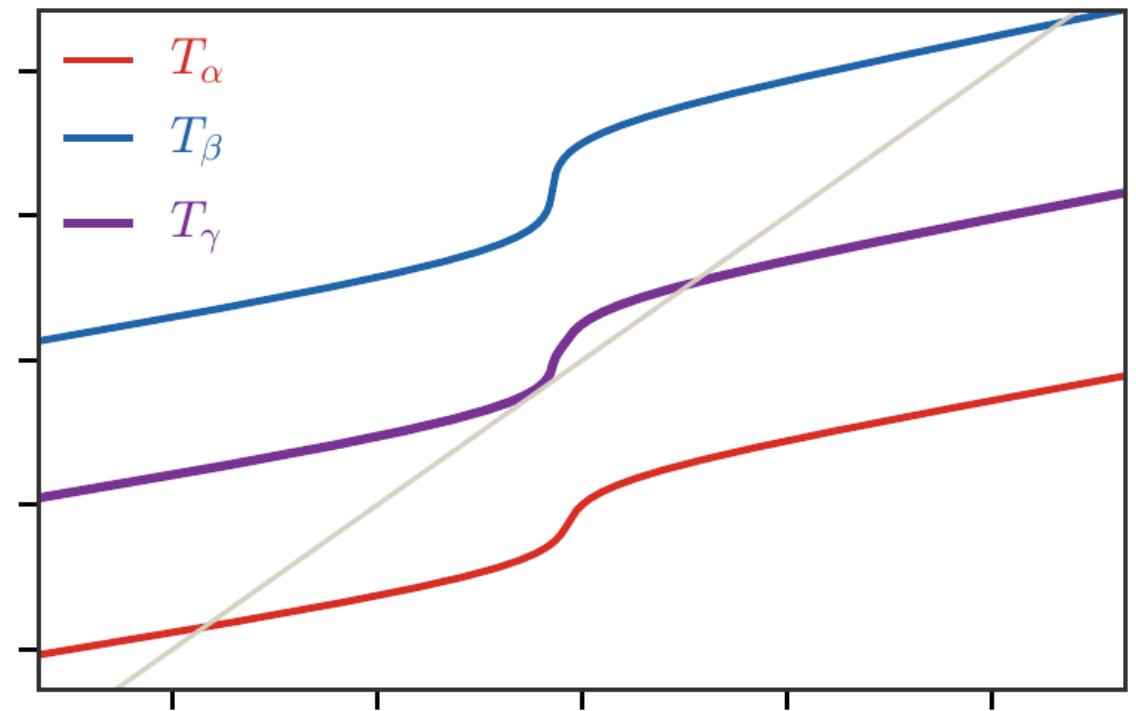
Semi-discrete OT appears when a continuous distribution is represented by a finite set of atoms:

- mesh generation and remeshing,
- color and texture processing,
- sampling and quantization,
- clustering with geometric costs,
- scalable approximations of continuous OT.

The computational object is a partition, not a dense coupling.

# Linearized OT preview

Dual and semi-discrete ideas also feed linearized OT: fix a reference measure and embed nearby measures by displacement fields.



# Live panel: linearized optimal transport

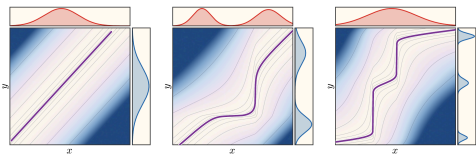


Linear OT turns transport maps into Euclidean coordinates for downstream analysis.

# Summary

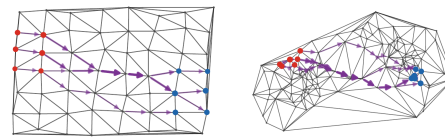
## What this deck establishes.

Duality turns transport into potentials. For  $W_1$ , these are Lipschitz witnesses. In the semi-discrete quadratic case, they are Laguerre weights defining a power diagram. Quantization and Lloyd algorithms reuse this geometry.



### Duality

certificates and gradients



### Wasserstein-1

Lipschitz witnesses



### Laguerre

cells and quantization

# Next deck

The next presentation focuses on dynamic optimal transport and flows over probability measures.

[Open deck 4: Dynamic Flows →](#)

